DEVELOPMENT OF A PHYSICAL FIDELITY-BASED PROBABILISTIC TSUNAMI HAZARD ASSESSMENT FRAMEWORK UTILIZING RANDOMIZED QUASI MONTE CARLO FOR SLIP DISTRIBUTION SAMPLING

SUBMITTED TO THE FACULTY OF THE
DEPARTMENT OF COMPUTER SCIENCE
COLLEGE OF INFORMATION TECHNOLOGY AND COMPUTING
BY

HENRICH MIGUEL DEL RIO CARPIO
CARL VINCE CALLAO DOMINGUEZ
JOHN MAR MAAGAD ESTIMADA
KARL ANDRE LOPEZ GUTIERREZ

IN PARTIAL FULFILLMENT OF THE
REQUIREMENTS FOR THE
DEGREE OF

BACHELOR OF SCIENCE IN COMPUTER SCIENCE

DECEMBER 2025

TABLE OF CONTENTS

Chapter I..... 1

INTRODUCTION.....1

1.2 Statement of the Problem..... 4
1.2.1 Slow Convergence on Earthquake Slip Distributions.....5
1.2.2 Uncertainty in Propagation Accuracy.....5
1.3 Research Objectives.....6
1.4 Scope and Limitations..... 7
1.5 Significance of the Study..... 8
1.6 Definition of Terms.....9
1.7 Proposed Probabilistic Tsunami Hazard Assessment Framework..... 10

Chapter II..... 11

REVIEW OF RELATED LITERATURE.....11

2.1 Tsunami Hazards and Hydrodynamic Complexity in the Philippine Trench..... 11
2.1.1 Recent Tsunamigenic Events (2012 vs. 2023).....13
2.1.2 Vulnerability of Coastal Infrastructure.....14
2.2 Probabilistic Tsunami Hazard Assessment (PTHA).....15
2.2.1 Treatment of Uncertainty..... 17
2.2.2 Integration with Seismic Hazard Models..... 18
2.2.3 Case Studies by Region.....18
2.2.4 Advances in PTHA Modeling..... 21
2.3 Gutenberg–Richter Law: Earthquake Frequency Distribution.....22
2.3.1 Application to Tsunami Hazard Analysis.....23
2.3.2 Tsunami Size–Frequency Relations.....25
2.3.3 Shallow vs. Deep Earthquakes in Tsunami Generation.....26
2.3.4 Critiques and Limitations of Gutenberg–Richter in Tsunami Context.....27
2.4 The Physics of Tsunami Generation: Coupled vs. Uncoupled Models. 28
2.4.1 Limitations of Superposition (Method 4).....29
2.4.2 Validation of the Fully Coupled Method (Method 1)..... 30
2.4.3 The Evolution to Dynamic Rupture-Tsunami Coupling..... 30
2.4.4 The Evolution to Dynamic Rupture-Tsunami Coupling..... 31

2.4.5 Application in Probabilistic Tsunami Hazard Analysis (PTHA)...32
2.4.6 Capturing Transient Wave Physics: Seismic and Acoustic Coupling.....32
2.5 Probabilistic Tsunami Hazard Assessment (PTHA) Paradigms.....33
2.6 Computational Efficiency in Hazard Modeling.....34
2.6.1 The Curse of Dimensionality and Monte Carlo Limitations.....34
2.6.2 Advanced Variance Reduction Strategies.....35
2.7 Randomized Quasi-Monte Carlo (RQMC) as a Methodological Solution 35
2.7.1 Mathematical Rigor and Statistical Guarantee.....36
2.7.2 Superior Convergence in High-Dimensional Spaces.....36

Chapter III.....38

METHODOLOGY.....38

3.1 Overall Methodology.....38
3.2 Preliminaries.....41
3.2.1 Methodological Foundations.....41
3.2.2 Collection of Fault Geometry and Seismotectonic Parameters.....42
3.3 Phase 1: Stochastic Slip Generation and Distribution sampling.....43
3.3.1 RQMC method.....43
3.3.2 Slip Random Field Model.....45
3.3.3 RQMC Sampling.....46
3.3.4 Seismic Moment Calculation and Generate-and-Scale Correction... 46
3.4 Phase 2: Probability Model.....48
3.4.1 Selection of the Gutenberg-Richter Recurrence Relation.....48
3.4.2 Seismicity Parameter Estimation for the Philippine Trench.....50
3.4.3 Integration into Hazard Aggregation.....52
3.5 Phase 3: High-Fidelity, Fully Coupled Tsunami Simulation.....53
3.5.1 Numerical Model: Modified GeoClaw.....53
3.5.2 Source Implementation.....54
3.6 Dual Validation Framework.....55
3.6.1 Statistical validation — convergence efficiency.....55
3.6.2 Physical validation — waveform fidelity.....55
3.7 Hazard Aggregation.....56
3.7.1 Annual Probability of Exceedance.....56
3.7.2 Outputs and Evaluation.....58
3.8 Experimental Procedure.....58

3.8.1 Simulation Campaign Structure.....58
3.8.2 Computational Workflow..... 59
3.8.3 Trial Execution and Replication..... 60
3.8.4 Experimental Setup and Computing Environment.....60
3.8.5 Software Implementation.....62
3.9 Evaluation Metrics..... 63
3.10 Initial Simulation..... 63

References:.....74

Chapter I
INTRODUCTION

The Philippine Trench, a north–south trending subduction zone extending over 1,300 kilometers along the eastern margin of the Philippine archipelago stands as one of the most seismically active yet critically understudied tsunamigenic sources in the western Pacific. Formed by the westward subduction of the Philippine Sea Plate beneath the Philippine Mobile Belt at rates exceeding 10 cm/year, this trench has repeatedly demonstrated its capacity to generate destructive earthquakes and tsunamis. Despite its clear hazard potential, systematic tsunami risk assessments for the Philippine Trench have been scarce, leaving densely populated coastal communities in the eastern Philippines dangerously unprepared. This knowledge gap has become increasingly untenable in light of recent seismic activity: two M_w 7.6 earthquakes, one in August 2012 and another in December 2023, both originated along this interface and produced measurable tsunamis with maximum tide gauge amplitudes of 3.7 cm and 12.5 cm, respectively (Ocean Engineering, 2025). Notably, the 2023 tsunami exhibited longer wave periods (6.7–28.2 minutes) compared to the 2012 event (8.0–18.3 minutes), a difference attributed to the deeper bathymetry surrounding the 2023 rupture zone, which allowed longer-wavelength energy to propagate more efficiently toward the coast (Ocean Engineering, 2025). These real-world events provide critical validation data while simultaneously underscoring the urgent need for advanced, probabilistic modeling frameworks that move beyond deterministic "what-if" scenarios toward quantifiable risk estimates grounded in physical realism and uncertainty quantification.

At the heart of tsunami generation lies the spatial distribution of coseismic slip along the fault plane, a primary source of uncertainty in hazard modeling. The initial sea surface displacement, which serves as the boundary condition for hydrodynamic simulations, is governed by the vertical seafloor deformation resulting from slip. Using the elastic dislocation theory of Okada (1985), this displacement at any surface point x = (x, y) can be expressed as

$$\eta_0(\mathbf{x}) = \sum_{i=1}^{N} \Delta u_z^{(i)}(\mathbf{x}; \boldsymbol{\theta}_i), \qquad \text{(Equation 1-1)}$$

where $\Delta u_z^{(i)}$ denotes the vertical displacement due to the i-th subfault and θi includes geometric and physical parameters most critically, slip magnitude s_i. Because slip cannot be observed prior to an earthquake and exhibits strong spatial heterogeneity due to frictional and stress variations

along the plate interface, it must be treated as a stochastic field. A common and physically motivated approach models log-slip as a Gaussian process:

$$\ln s(\mathbf{r}) \sim \mathcal{GP}(\mu_s, \mathcal{C}(\|\mathbf{r} - \mathbf{r}'\|; \ell)), \qquad \text{(Equation 1-2)}$$

where μ_s is the mean log-slip, $C(\cdot;\ell)$ is a covariance function (e.g., exponential or Matérn), and ℓ is the correlation length that controls the spatial smoothness of slip patches (Goda & Song, 2016). This formulation captures the physical reality that large slip tends to cluster in asperities, while surrounding regions experience little or no displacement. The total seismic moment M_0 provides a crucial physical constraint that couples all slip values:

$$M_0 = \mu \int_\Omega s(\mathbf{r})\, dA = \mu \bar{s} A, \qquad \text{(Equation 1-3)}$$

where Ω is the rupture area, $\bar{S}$ is the spatial mean slip, $A=|\Omega|$, and μ is the effective shear modulus. For a target magnitude M_w, M_0 is fixed via the moment–magnitude relation: $M_w = \frac{2}{3} \log_{10}(M_0) - 10.7$ (Hanks & Kanamori, 1979). Thus, while slip heterogeneity is permitted, it must collectively satisfy this global moment budget, a nonlinear constraint that renders the sampling problem high-dimensional and nontrivial.

To explore this uncertainty space efficiently, traditional Monte Carlo (MC) methods generate independent slip realizations and compute corresponding tsunami metrics, but their slow convergence rate ($O(N^{-1/2})$) makes them computationally prohibitive for expensive hydrodynamic simulations. Quasi-Monte Carlo (QMC) methods, which use low-discrepancy sequences such as Sobol' (Sobol', 1967), achieve faster convergence often close to $O(N^{-1})$ in moderate dimensions but lack built-in variance estimation, limiting their utility for risk-informed decisions. This limitation is overcome by Randomized Quasi-Monte Carlo (RQMC), which applies random scrambling to a QMC sequence to produce R independent replicates, enabling unbiased estimation and robust error quantification (L'Ecuyer & Lemieux, 2002). The RQMC estimator for the expected maximum tsunami height at location x becomes

$$\hat{\mu}_{\mathrm{RQMC}}(\mathbf{x}) = \frac{1}{R} \sum_{r=1}^{R} \left(\frac{1}{N} \sum_{k=1}^{N} H_{\max}^{(r,k)}(\mathbf{x}) \right), \qquad \text{(Equation 1-4)}$$

with sample variance that supports confidence intervals essential for actionable risk metrics. This framework enables statistically rigorous,

computationally efficient Probabilistic Tsunami Hazard Analysis (PTHA), a capability urgently needed for the Philippine Trench (Ocean Engineering, 2025).

To accurately simulate tsunami generation and propagation, this study adopts Method 1: the fully-coupled modeling approach, which simultaneously solves earthquake rupture dynamics, seismic/acoustic wave propagation through the solid Earth and ocean, and tsunami hydrodynamics across the basin and nearshore zones. This method ensures physical consistency by dynamically linking the evolving slip distribution s ($\mathbf{r}$,t) to both seismic radiation and seafloor deformation, thereby avoiding the artificial decoupling inherent in traditional two-step models (e.g., static Okada displacement followed by shallow-water propagation). The system is governed by coupled partial differential equations, including the elastic wave equation for crustal motion and the nonlinear shallow-water equations for tsunami evolution (LeVeque et al., 2011). While this approach offers superior physical fidelity, its predictive accuracy particularly in coastal inundation depends critically on the numerical treatment of the propagation phase.

Recognizing this, this study proposes a key enhancement to Fully-coupled Method: the implementation of high-order, adaptive, and well-balanced numerical schemes to improve propagation accuracy. Standard implementations often rely on second-order finite volume methods on uniform grids, which can introduce numerical dispersion and dissipation especially in shallow coastal zones with complex bathymetry. To address this, this study integrates third-order Weighted Essentially Non-Oscillatory (WENO) reconstruction to preserve wavefront sharpness, bathymetry-aware adaptive mesh refinement (AMR) to dynamically increase resolution near shorelines and steep gradients, and well-balanced fluxes to maintain the lake-at-rest condition over irregular topography (George, 2008; LeVeque et al., 2011). These enhancements ensure that small but critical features such as run-up height, inundation extent, and arrival time are captured with greater fidelity, directly impacting risk estimates for vulnerable communities.

This improved framework will be applied specifically to the Philippine Trench, building directly on the scenarios outlined in the 2025 study. That work modeled both the 2012 and 2023 M_w 7.6 events and expanded to hypothetical M_w 8.0, 8.5, and 9.0 ruptures, revealing that a worst-case M_w 9.0 earthquake could generate coastal tsunami waves up to 17.4 meters high comparable to the 2011 Tohoku and 2004 Indian Ocean tsunamis (Ocean Engineering, 2025). Even an M_w 8.5 event could inundate 218 buildings in the coastal town of Dapa, including a designated evacuation center, thereby compromising the very infrastructure meant to ensure public safety (Ocean Engineering, 2025). This study extends this deterministic analysis by embedding it within an RQMC-based probabilistic framework that samples

thousands of physically consistent slip distributions, propagates each through the enhanced fully-coupled model, and quantifies exceedance probabilities for tsunami height, inundation depth, and building exposure.

The annual probability that tsunami height exceeds a threshold h at location x is given by

$$P(H(\mathbf{x}) > h) = \int_{m_{\min}}^{m_{\max}} \nu(m) \cdot P(H(\mathbf{x}) > h \mid M = m)\, dm \qquad \text{(Equation 1-5)}$$

where $\nu(m)=\beta \exp(-\beta m)$ is the Gutenberg–Richter frequency–magnitude distribution, and the conditional probability is estimated via RQMC using an indicator function over the ensemble of simulations (González et al., 2009). This formulation produces tsunami hazard curves and inundation probability maps that can directly inform land-use planning, early warning protocols, and evacuation strategies in the Philippines, a nation where over 60% of the population lives in coastal zones and institutional capacity for tsunami preparedness remains limited.

In conclusion, the Philippine Trench is no longer a hidden hazard but a quantifiable, high-consequence threat demanding advanced scientific attention. By integrating stochastic slip physics, efficient RQMC sampling, and a propagation-enhanced fully-coupled tsunami model, this thesis responds directly to the urgent call from the 2025 study for comprehensive tsunami risk assessment along this underexplored subduction zone (Ocean Engineering, 2025). The resulting framework not only advances methodological rigor but also delivers actionable insights for building true resilience in one of the world's most disaster-prone archipelagos.

1.2 Statement of the Problem

The Philippine Trench remains significantly understudied in terms of tsunami hazard projection, leaving substantial gaps in understanding its potential impacts (Heidarzadeh et al., 2025). While existing modeling efforts (e.g., Reyes et al., 2020) provide a useful foundation, closer examination reveals persistent limitations: uncertainties in earthquake source characterization (Qiu et al., 2019) and trade-offs in numerical methods, such as simplified wave-propagation models used to maintain computational efficiency (FUNMAP study, 2025). These technical shortcomings can hinder realistic simulation, particularly in capturing complex wave behavior and achieving numerical convergence. Therefore, this study aims to refine and

enhance current models to overcome these limitations and provide more reliable, physically realistic simulations tailored to the Philippine Trench.

1.2.1 Slow Convergence on Earthquake Slip Distributions

Probabilistic tsunami hazard assessment (PTHA) requires the generation of a large ensemble of stochastic earthquake slip distributions to account for epistemic uncertainties in seismic source modeling. These slip realizations are used to simulate tsunami wave heights at coastal locations, forming the basis for risk-informed planning and mitigation. Traditionally, the Monte Carlo (MC) method has been employed to generate these stochastic slip scenarios due to its simplicity and general applicability. However, MC is known to suffer from slow convergence rates, specifically $O(N^{-1/2})$, which significantly increases the computational cost required to achieve stable and reliable hazard estimates (Hou et al., 2019).

As demonstrated in convergence analyses of tsunami simulations, thousands to tens of thousands of samples may be needed for each magnitude bin to stabilize maximum coastal tsunami height distributions (e.g., via the coefficient of variation). This demand on sample size makes MC-based modeling inefficient, especially in high-resolution, fully coupled tsunami models. Therefore, there is a need to explore alternative sampling methods that can achieve more accurate results with fewer simulations.

1.2.2 Uncertainty in Propagation Accuracy

Maintaining Physical Fidelity Against Simplifications: Simpler tsunami models (which rely on shortcuts and approximations) fail when simulating wave travel in deep water because they neglect phenomena like wave dispersion (waves changing speed and shape as they travel) and non-hydrostatic filtering (how deep water naturally filters out small-scale wave features created by the seafloor movement) (Abrahams et al., 2023).When these effects are ignored, predictions suffer major errors: the simpler models can "overpredict the tsunami wave amplitude by a factor of two" (Abrahams et al., 2023) or incorrectly predict a much larger initial wave that "arrives too early" (Abrahams et al., 2023). This study must ensure advanced simulation which correctly includes these dispersive and filtering effects maintains this accuracy as the wave propagates over distance.

1.3 Research Objectives

This study aims to develop a high-fidelity probabilistic tsunami hazard assessment framework specifically for the Philippine Trench. The objectives are as follows:

1. To implement a Randomized Quasi-Monte Carlo (RQMC) sampling model that efficiently generates synthetic earthquake slip distributions.

2. To enhance the propagation accuracy of the Fully Coupled Method (Method 1) explicitly selected over the Superposition Method (Method 4) to strictly resolve non-hydrostatic filtering effects that otherwise cause twofold wave amplitude overprediction by integrating high-order numerical schemes, and to apply this framework together with the RQMC sampling model to the Philippine Trench.

3. To design and implement a functional prototype that integrates RQMC sampling into the Enhanced Fully-Coupled Method, structured to meet established PTHA hazard evaluation frameworks.

4. To evaluate the effectiveness of the Enhanced Fully Coupled Method with RQMC sampling using statistical and numerical standards grounded in NTHMP tsunami validation benchmarks and established PTHA hazard evaluation frameworks.

1.4 Scope and Limitations

Scope

This study focuses exclusively on the Philippine Trench as the primary tsunamigenic source, modeling various stochastic rupture scenarios to assess hazards along the eastern seaboard of the Philippines. The geographic area of interest specifically targets high-risk coastal zones in the Caraga and Davao regions, which are geographically exposed to near-field tsunamis generated by the subduction interface. The modeling framework integrates a Randomized Quasi-Monte Carlo (RQMC) sampling technique to efficiently generate synthetic earthquake slip distributions, addressing the need for robust uncertainty quantification in probabilistic assessments. These stochastic sources drive a fully-coupled tsunami generation model (Method 1), which simultaneously solves the elastic and hydrodynamic equations to capture dynamic rupture effects often missed by static models. To ensure physical fidelity during the propagation phase, the study employs enhanced numerical schemes, specifically third-order Weighted Essentially Non-Oscillatory (WENO) reconstruction and Adaptive Mesh Refinement (AMR), to resolve complex wave interactions over the Philippine Trench bathymetry. The resulting hazard assessment quantifies critical metrics including tsunami wave height, arrival time, inundation depth, and structural exposure, with the physical accuracy of the model validated against observational data from the 2012 and 2023 M_w 7.6 tsunamigenic events.

Limitations

Despite the methodological enhancements, this study operates under specific technical and physical constraints. First, the accuracy of coastal inundation estimates is limited by the resolution of bathymetric and topographic data; the model relies partly on global datasets like GEBCO, which may not fully resolve fine-scale nearshore features such as coral reefs or small man-made structures that influence local run-up behavior. Second, regarding source complexity, the study assumes megathrust-only ruptures and does not account for secondary tsunamigenic sources such as splay faults, outer-rise earthquakes, or submarine landslides, which could act as hazard amplifiers in specific local scenarios. Third, while the use of RQMC significantly reduces computational cost compared to traditional methods, the ensemble size is still constrained by available high-performance computing resources, which may limit the precise estimation of tail-risks for extremely rare, worst-case M_w 9.0 events. Finally, the assessment of risk is strictly geophysical and structural; while building exposure is quantified, complex

socioeconomic factors such as population dynamics, real-time evacuation behavior, and economic loss modeling are beyond the scope of this hydrodynamic simulation study.

1.5 Significance of the Study

The primary academic contribution of this research is the development of a high-fidelity, fully coupled simulation framework that resolves the "computational bottleneck" inherent in Probabilistic Tsunami Hazard Analysis (PTHA). Existing studies often rely on simplified static dislocation models to save computational time, but recent research indicates that such decouplings can lead to significant errors in predicting wave amplitude and arrival times. By integrating Randomized Quasi-Monte Carlo (RQMC) sampling, this study provides a methodological breakthrough, demonstrating that physically realistic slip heterogeneity can be modeled with superior convergence rates compared to traditional Monte Carlo methods. This allows for the feasible application of the computationally expensive Fully Coupled Method (Method 1) within a probabilistic framework, setting a new benchmark for accuracy in hydrodynamic modeling. Furthermore, this research fills a specific gap in the literature by applying these advanced variance-reduction techniques to the Philippine Trench, a subduction zone that remains understudied compared to the Japan or Nankai trenches despite its potential for generating destructive earthquakes.

For the Philippines, specifically the vulnerable communities along the eastern seaboard where institutional tsunami preparedness is often limited, this study provides actionable, quantitative data to enhance disaster resilience. Moving beyond deterministic "worst-case" scenarios, the probabilistic outputs such as inundation probability maps and hazard curves offer a scientific basis for refining Local Government Unit (LGU) land-use plans, such as restricting the construction of critical infrastructure in statistically high-risk coastal zones. The enhanced accuracy of the propagation model directly supports the calibration of early warning systems by establishing site-specific alert thresholds based on physically validated wave arrival times rather than generic estimates. Additionally, the study informs evacuation planning by identifying areas where designated evacuation centers may be compromised under specific magnitude scenarios, necessitating a re-evaluation of safety protocols for coastal populations. Ultimately, this framework contributes technical evidence to support national policy updates, including the Philippine Development Plan and the National Disaster Risk Reduction and Management Framework, by offering a replicable blueprint for science-based risk assessment in data-scarce, high-hazard regions.

1.6 Definition of Terms

Philippine Trench: A north–south trending subduction zone along the eastern margin of the Philippines, where the Philippine Sea Plate subducts westward beneath the Philippine Mobile Belt at rates exceeding 10 cm/year.

Fully Coupled Method: A high-fidelity tsunami modeling approach that simultaneously solves the equations of elastodynamics (for seismic wave propagation in the solid Earth) and hydrodynamics (for tsunami wave propagation in the ocean), capturing the complete multiphysics wavefield.

Stochastic Slip Distribution: A spatially variable model of earthquake slip treated as a random field, typically log-normal with spatial correlation, used to represent uncertainty in rupture heterogeneity.

Randomized Quasi-Monte Carlo (RQMC): An advanced sampling technique that combines low-discrepancy sequences with randomization to achieve faster convergence and variance estimation in high-dimensional uncertainty quantification.

Propagation Accuracy: The fidelity with which a tsunami model simulates wave evolution from deep ocean to coast, particularly in resolving inundation extent, run-up height, and arrival time enhanced in this study via WENO, AMR, and well-balanced numerics.

Probabilistic Tsunami Hazard Analysis (PTHA): A framework that quantifies the likelihood of tsunami impacts (e.g., wave height exceeding a threshold) over time, accounting for uncertainties in earthquake source, path, and site effects.

M_w (Moment Magnitude): A logarithmic scale measuring the total energy released by an earthquake, based on seismic moment M_0 , defined as $M_w = 3/2 \ \log_{10}(M_0) - 10.7$.

1.7 Proposed Probabilistic Tsunami Hazard Assessment Framework

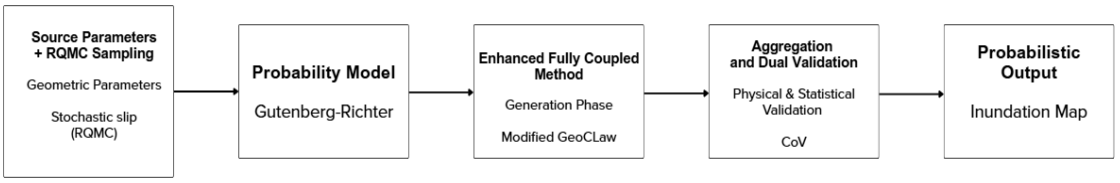


Figure 1.1. Proposed Probabilistic Tsunami Hazard Assessment Framework

This study proposes a Physical Fidelity-Based Probabilistic Tsunami Hazard Assessment framework designed to rigorously evaluate tsunami risks. The methodology commences with the definition of source parameters, utilizing Randomized Quasi-Monte Carlo (RQMC) sampling to construct a robust probability model that efficiently covers the uncertainty space. These probabilistic inputs drive the Generation and Propagation phase, which applies "Enhanced Method 1" to simulate wave dynamics. Crucially, a dual validation mechanism is integrated at this stage to verify the physical accuracy of the simulations before the results are finalized. Ultimately, the validated outputs are synthesized through an aggregation process to quantify the comprehensive hazard levels.

Chapter II
REVIEW OF RELATED LITERATURE

2.1 Tsunami Hazards and Hydrodynamic Complexity in the Philippine Trench

The Philippine Trench represents one of the most seismically active yet complex subduction zones in the western Pacific. While historical hazard assessments often utilized static worst-case scenarios, recent geological events have necessitated a shift toward more nuanced, physically validated models.

The geodynamic complexity of the Philippine Trench significantly influences its seismic and potential tsunami behavior. Lallemand et al. (1998) demonstrate that the trench is divided into distinct segments, with the deepest portion (~10,100 m) located near 9–10°N, coinciding with the maximum depth of the subducting Philippine Sea Plate (~200 km) and the oldest arc volcanism (~2.5 Ma). South of 6°N, the trench shallows rapidly, interplate seismicity diminishes, and subduction appears to slow down due to interaction with the Molucca Sea slab and the Sangihe subduction system. The authors interpret this as evidence of decreased plate coupling in the southern segment, which may reduce the likelihood of large thrust earthquakes—and consequently, major tsunamis—in that region. In contrast, the central segment (7–10°N) exhibits stronger mechanical coupling, deeper subduction, and more pronounced bending-related faulting, suggesting higher seismic and tsunami potential. These tectonic insights are crucial for regional tsunami hazard assessments, as they highlight spatial heterogeneity in subduction behavior along the Philippine Trench.

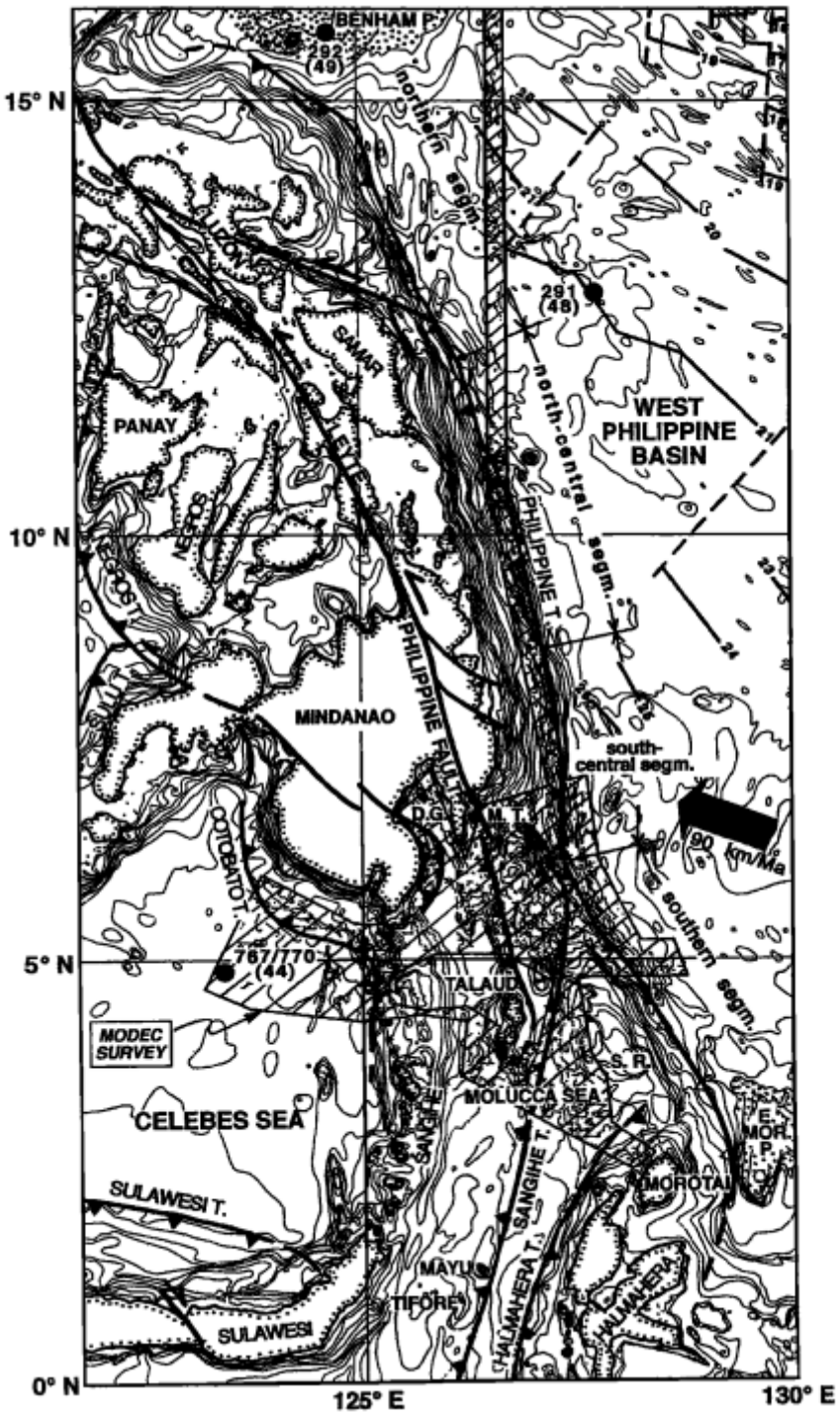


Figure 2.1. Tectonic map of the Philippine and Sangihe Trenches

High-resolution P-wave tomography across the southern Philippine Trench documents pronounced structural complexity with three subducting slabs (Celebes, Molucca, and Philippine Sea) and an overturned geometry of the Philippine Sea slab reaching depths of ~450–600 km. Fan & Zhao (2018) invert a large dataset of local and teleseismic arrivals to obtain images with

lateral resolution ≈0.6° and show that the southern segment likely initiated subduction at ~20–25 Ma, with a later subduction reorganization about ~5 Ma associated with the closure of the Molucca Sea. The resulting slab segmentation, depth variations, and lateral heterogeneities inferred from their tomographic cross-sections suggest spatially variable seismic coupling and fault behavior along the trench conditions that can produce segmented or multifocal tsunami sources and complicate near-field hydrodynamic responses. Therefore, tomographic constraints such as those presented by Fan & Zhao provide essential geophysical context for tsunami hazard models that incorporate segmented rupture scenarios, variable slip distributions, and site-specific bathymetry.

The spatiotemporal evolution of subduction along the Philippine Trench has direct implications for tsunami potential. Ozawa et al. (2004) used K–Ar dating of volcanic rocks along the Philippine Volcanic Arc to demonstrate that subduction initiated ~8 Ma in the north (Bicol Peninsula) and propagated southward, reaching Leyte by ~3.5 Ma and likely extending further south more recently. This north-to-south progression aligns with geophysical evidence showing a systematic decrease in trench depth, slab length, and interplate coupling toward the south. Consequently, the central and northern segments (e.g., 7°–12°N) are tectonically mature and capable of generating large megathrust earthquakes, whereas the southern segment (south of 6°N) remains immature and weakly coupled, reducing its tsunami hazard potential. This segmentation underscores the need for regionally differentiated tsunami risk models along the Philippine Trench.

2.1.1 Recent Tsunamigenic Events (2012 vs. 2023)

The urgency of updating hazard models for this region is underscored by recent empirical data. Heidarzadeh et al. (2025) performed a comparative analysis of two major earthquakes (M_w 7.6) that occurred along the trench in August 2012 and December 2023. Their study revealed that despite having identical magnitudes, the resulting tsunamis exhibited drastically different hydrodynamic behaviors due to the rupture depth and bathymetry. The 2023 event produced waves with significantly longer periods (6.7–28.2 minutes) compared to the 2012 event (8.0–18.3 minutes), a phenomenon attributed to the deeper water column at the 2023 source. This finding supports the argument that simple magnitude-based projections are insufficient; hazard models must account for the spatial heterogeneity of the rupture to capture these complex wave characteristics.

2.1.2 Vulnerability of Coastal Infrastructure

The gap between scientific understanding and infrastructure readiness remains wide. Simulation data from the 2025 study highlighted that an M_w 8.5 event well within the trench's capacity would likely inundate critical facilities, including the designated evacuation center in the coastal town of Dapa. This exposure of "safe zones" indicates a failure in current deterministic planning maps, validating the need for the probabilistic inundation mapping proposed in this study.

Geomorphic evidence from uplifted marine terraces along eastern Mindanao provides critical insight into the tsunami potential of the southern Philippine Trench. Ramos et al. (2012) identified up to four Holocene terrace levels (1–12 m above sea level) along the Davao Oriental coast, with radiocarbon ages between 8,080 and 4,140 cal yr BP. The staircase morphology and 1–3 m terrace risers are interpreted as the result of repeated coseismic uplift events, likely caused by large to great ($M \geq 8$) megathrust earthquakes along the Philippine Trench. Notably, historical earthquakes in the same region (e.g., the 1992 M_w 7.1–7.2 events) generated a 5-meter tsunami but no detectable coastal uplift, suggesting that the terraces record significantly larger prehistoric ruptures. This implies that the southern Philippine Trench, though tectonically younger and less seismically active than its central counterparts remains capable of generating tsunamigenic earthquakes, challenging assumptions of uniformly low hazard in this segment.

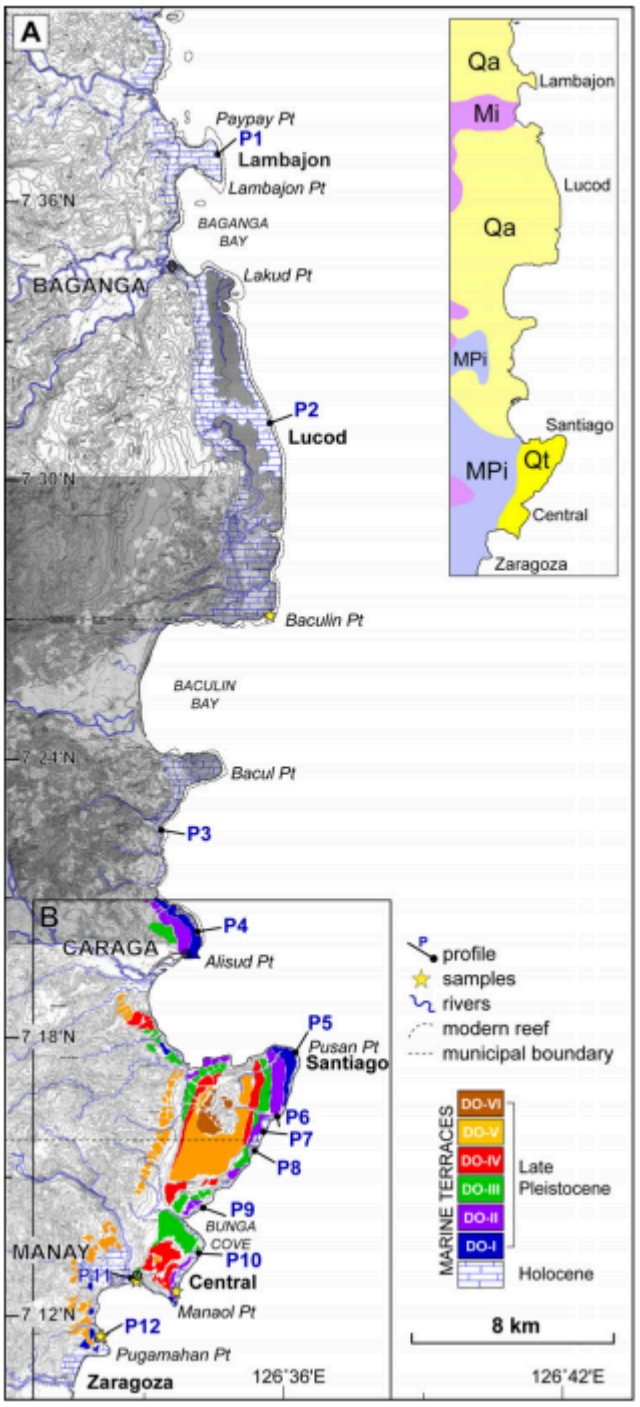


Figure 2. 2. Map of uplifted marine terraces in Davao Oriental

2.2 Probabilistic Tsunami Hazard Assessment (PTHA)

Probabilistic Tsunami Hazard Assessment (PTHA) applies formal probabilistic methods (adapted from seismic hazard analysis) to quantify the likelihood of tsunami waves of various sizes. In PTHA, all potential tsunami sources and their occurrence rates are defined, tsunamis are simulated for

each scenario, and the results are combined into exceedance-probability curves or maps. The intensity measure (IM) is a physical tsunami metric (e.g. coastal wave amplitude or run-up height). PTHA differs from deterministic "worst-case" approaches by assigning probabilities to scenarios, yielding hazard curves such as the annual probability of exceeding a given run-up. Formally, PTHA estimates the mean annual rate $\lambda(IM \geq im)$ that a tsunami intensity IM at a location exceeds threshold im . This is done by integrating tsunami effects (from numerical models) with seismic event rates (from magnitude–frequency models). In practice, PTHA proceeds in three core stages:

Seismic Source Modeling: Define all tsunamigenic sources (e.g. subduction megathrusts, crustal faults, submarine slides, volcanoes) and assign earthquake occurrence rates. Sources may be zoned by tectonic region or fault, and each is given a magnitude–frequency distribution (commonly a Gutenberg–Richter (G–R) law or a "characteristic" model). The seismicity model (Poissonian or renewal) yields the annual rate of each earthquake magnitude. Logic trees or multiple branches are often used to capture epistemic uncertainty in parameters (e.g. different b-values or maximum magnitudes). For example, Sorensen et al. (2012) applied truncated G–R relations for Mediterranean source zones by fitting catalog data, while Baba et al. (2022) used a G–R model for Nankai Trough earthquake rates.

Tsunami Simulation: For each seismic scenario (defined by location, fault geometry, magnitude, slip distribution, etc.), compute the resulting tsunami using numerical wave-propagation models (shallow-water or Boussinesq models with nested grids). High-resolution inundation models are often used nearshore. Many studies simulate seafloor displacement and wave evolution for each event. The output is an intensity measure (wave height, flow depth) at coastal sites. For example, Gonzalez et al. (2009) developed the first PTHA inundation maps by coupling PSHA with inundation modeling for Seaside, Oregon. They computed 100- and 500-year tsunami amplitudes (1% and 0.2% annual exceedance) from both far-field (Alaska) and near-field (Cascadia) earthquakes.

Statistical Hazard Estimation: Combine the tsunami outputs with event rates to build hazard curves. Each earthquake scenario has a weight (its annual rate) from the seismic model, and produces a maximum IM at the site. By summing over all scenarios (e.g. via Monte Carlo or numerical integration), one obtains the probability that IM exceeds given levels in a time window. The result can be expressed as a curve of excess probability vs. IM, or as maps of IM at specified return periods. UNESCO guidelines describe PTHA as first "establishing parameters for all possible tsunami sources and occurrence rates" then simulating and aggregating results into exceedance curves. In practice, PTHA often uses Monte Carlo: e.g. Geist and Parsons

(2006) randomized hypocenter and slip to build a tsunami amplitude–probability curve.

2.2.1 Treatment of Uncertainty

PTHA explicitly quantifies uncertainty. Uncertainties are usually divided into epistemic (due to lack of knowledge or competing models) and aleatory (intrinsic randomness). Epistemic uncertainties (e.g. which seismic source model is correct, how to represent fault geometry) are handled via logic trees or ensembles of alternative assumptions, each branch weighted by credibility. Aleatory variability (e.g. random slip heterogeneity, tide level, simulation error) is treated statistically by probability density functions. Common examples include:

Logic Trees: Many PTHAs use logic trees to capture uncertainty in source characterization and magnitude-frequency. For instance, alternative Gutenberg–Richter b-values or corner magnitudes might be assigned weights, and the final hazard curve is averaged over branches. Geist and Parsons (2005) compared "characteristic" vs. G–R magnitude distributions for Cascadia by Monte Carlo, illustrating how different recurrence models affect hazard.

Slip Variability: The spatial distribution of fault slip in an earthquake can strongly affect tsunami size. Modern PTHAs often generate multiple slip realizations per rupture to sample aleatory slip variability. For example, Mazet-Roux et al. (2019) modeled each rupture area with five stochastic slip distributions based on overlapping circular "asperities," enforcing a k^(-2) decay of slip spectrum. This approach captures the natural heterogeneity of slip and its effect on tsunami generation.

Tidal and Sea-Level Uncertainty: Background water level (tide or sea level) adds noise to run-up predictions. PTHAs incorporate this by sampling over tidal phases or adding a distribution of sealevel offsets. Baba et al. (2022) explicitly included astronomical tide variations in their hazard curves. The U.S. Crescent City study used a "pattern method" to model tidal uncertainty when mapping inundation. More generally, stochastic sampling of tide phase and projected sea-level rise is becoming standard in recent assessments.

Model and Measurement Uncertainty: Differences between modeled and observed tsunami heights introduce additional aleatory spread. Davies et al. (2017) noted that model observation mismatches were larger than previously assumed, implying that failure to include this uncertainty underestimates hazard. They found that explicitly accounting for such deviations increases predicted run-ups for a given exceedance probability.

In all cases, uncertainties are propagated through to the final hazard (e.g. via Monte Carlo or convolution). The output is typically given with confidence bounds (from logic-tree weighting) on the hazard curves or maps. For example, UNESCO guidelines remark that PTHA "explicitly addresses different types and sources of uncertainty" by developing and weighting alternative models.

2.2.2 Integration with Seismic Hazard Models

PTHA is closely linked with seismic hazard assessment. The earthquake catalogs and frequency models used in PTHA usually come from PSHA-type analysis. Magnitude–frequency parameters (b-value, corner magnitude) and source zonations from seismic studies are adopted or adjusted for tsunami application. For example, many PTHAs use Gutenberg–Richter fits to regional seismicity as the basis for earthquake occurrence rates. In the Mediterranean PTHA, Sørensen et al. fitted a truncated G–R law per source zone using historical earthquake catalogs. Similarly, Baba et al. based Nankai Trough earthquake probabilities on a G–R model. Alternative seismic hazard branches (e.g. using different b-values or a characteristic event assumption) can be included in the logic tree. Thus, PTHA builds on PSHA methods: earthquake event rates ($\lambda(M)$) feed directly into tsunami event rates, and Poissonian or time-dependent occurrence models from PSHA carry over. The strong coupling means that any update in seismic hazard (new sources, revised recurrence) will directly affect tsunami hazard estimates.

2.2.3 Case Studies by Region

Probabilistic tsunami hazard studies have been carried out worldwide, ranging from local to basin scales. Representative examples include:

Japan – Nankai Trough: Baba et al. (2022) applied PTHA to eastern Shikoku, Japan. They generated 3480 rupture scenarios of past Nankai earthquakes and computed corresponding tsunamis. Earthquake probabilities were based on a Gutenberg–Richter model, and tidal variation was included. The resulting hazard curves (exceedance vs. wave height) agreed well with historical tsunami frequencies. Sensitivity tests revealed that slip heterogeneity and coastal defenses significantly affect hazard levels. The study also produced high-resolution inundation maps with probabilistic contours.

Pacific Northwest (Cascadia, USA): Gonzàlez et al. (2009) performed a PTHA for Seaside, Oregon (Cascade Range region). They combined tsunami inundation modeling with PSHA-derived earthquake rates to produce maps of 100- and 500-year tsunami amplitudes. They found that 100- year tsunamis are usually from far-field sources (Alaska–Aleutians) with <4 m amplitudes, whereas 500-year events are dominated by local Cascadia megathrust ruptures (>10 m). The main uncertainties were in earthquake recurrence intervals, sea level, and bathymetry. Similar PTHAs have been conducted for Crescent City, California, with updated source sets and tidal uncertainty treatment, confirming that 100- and 500-year inundation extents are very sensitive to seismic-source assumptions

Mediterranean Sea: Sørensen et al. (2012) presented the first basin-wide PTHA for the Mediterranean. Using Monte Carlo simulation with zone-specific G–R seismicity models, they estimated the annual probability of exceeding a given tsunami amplitude at any coast. The highest hazard was along the Hellenic Arc (Greece), but almost all coastlines showed non-negligible risk. Notably, they estimated that the probability of anywhere in the Mediterranean experiencing a tsunami ≥1 m in the next 30 years is nearly 100%. Their method identifies the dominant source (region) for hazard at each location and can inform warning systems.

Global (Earthquake-Generated Tsunamis):Davies et al. (2017) performed a global PTHA covering all major subduction zones worldwide. They considered only earthquake sources (≈80% of historical tsunamis) and generated global maps of tsunami run-up for specified exceedance rates. This work highlighted that large uncertainties (especially in the frequency of great earthquakes) produce large spread in predicted run-ups. They also found that accounting for modeling errors increases predicted hazard: models tended to under-predict historical run-ups, implying a higher real hazard than naive projections.

Other Regions: PTHA has also been done for many other areas. For example, Tang et al. (2017) assessed PTHA for Southeast China/Taiwan, Hokkaido (Japan), Alaska, and several national studies in Chile and Indonesia. In each case, local seismicity and source faults are combined with tsunami simulation to produce site-specific hazard curves. While this study cannot list all studies, the general pattern is similar: evaluate a suite of earthquakes (with varied magnitudes and slips), run tsunami models, and compute exceedance probabilities.

Table 2.1. Tsunami Modeled sources Comparison

REGION / STUDY	TSUNAMI SOURCES MODELED	KEY FINDINGS
Japan (Nankai) – Baba et al. 2022	Nankai subduction M>9 | earthquakes	3480 scenarios, G-R rates, tidal sampling; hazard curves agree with historic tsunami frequency: slip heterogeneity and defenses matter.
Cascadia (Oregon, USA) - Gonzalez et al. 2009	Cascadia M=9 megathrust & Alaska-Aleutians events	100-yr tsunami mainly from far-field (Alaska, <4m); 500-yr from local Cascadia (>10m); mapped 1%/0.2% contours; main uncertainties in recurrence and sea-level.
Mediterranean – Sørensen et al. 2012	Hellenic Arc + other Med quakes	Monte Carlo with G-R zones; ~100% chance of ≥1m wave in 30 yr somewhere; highest hazard east Med; identifies dominant sources for each coast.
Global – Davies et al. 2017	All major subduction zones	Earthquake-only PTHA; global run-up hazard maps found very large epistemic uncertainties; accounting model-data discrepancy raises predicted hazard.
California (Crescent City) – Gonzalez et al. 2013	Cascadia and distant sources	Improved PTHA with more sources and tidal modeling; 100/500 yr hazard maps; found high sensitivity of maps to parameter choices (recommended p-contour mapping).

Sources: Baumgart et al. 2025; Rawizza et al. 2020; Cassadas et al. 2023; Roderiyue

2.2.4 Advances in PTHA Modeling

Recent developments are expanding and accelerating PTHA:

Machine-Learning Surrogates: High-fidelity tsunami simulations are computationally intensive, especially when thousands of scenarios are needed. New studies train ML models to predict tsunami impacts. For example, Ramalingam et al. (2025) ran hundreds of tsunami simulations for Tōhoku subduction earthquakes and trained a variational encoder–decoder network to predict coastal waveforms and maximum inundation depths from a small number of deep-ocean input values. The ML surrogate reproduced the full-physics results with high accuracy, enabling rapid hazard mapping along long coastlines with limited simulation runs. Such surrogate models (including neural nets and Gaussian-process emulators) show great promise for accelerating PTHA.

Stochastic Rupture Models: Better representation of slip variability improves realism. Recent PTHA studies use stochastic rupture generators (e.g. self-similar spectral models) to sample many possible slip patterns. For instance, Mazet‑Roux et al. (2019) generated five random slip distributions per rupture by superimposing circular sub‑faults ("asperities") to enforce a k^(-2) spectral decay. This captures aleatory slip heterogeneity in each scenario. Similar approaches (random kinematic slips) allow PTHA to automatically include slip randomness instead of assuming a uniform slip on the fault.

Multi‑Hazard ("Hybrid") Frameworks: PTHA is evolving into a comprehensive multi-source framework. Grezio et al. (2021) discuss "Total PTHA (TotPTHA)" methods that merge multiple tsunamigenic sources earthquakes, submarine slides, volcanic blasts, even meteorological or anthropogenic triggers into a single probabilistic assessment . The TotPTHA formalism accounts for source interactions (e.g. an earthquake triggering a landslide) and sums over all source types to compute a total tsunami hazard. In essence, each source class contributes its own PTHA, and these are combined (with possible conditional probabilities) to get the overall hazard. While still emerging, this approach reflects the growing recognition that non-seismic tsunamis (landslides, volcanoes) can be important locally, and that all must be considered for a "complete" hazard analysis.

Statistical Emulation and Fast Schemes: Beyond deep learning, simpler surrogate approaches are also used. Emulators built from regression or machine-learning reduce the number of required full simulations. The "Pattern Method" and other statistical schemes address specifically the distribution of tidal stages or other factors. Such methods are being incorporated into PTHA to speed up computations while quantifying uncertainty.

Time-Dependent Models: Traditional PTHA assumes Poisson (time-invariant) occurrence, but some research is exploring renewal models. For subduction zones with quasi-periodic great quakes (e.g. Cascadia), a time-dependent hazard analysis can modulate probabilities based on elapsed time since the last event. These approaches borrow from seismic renewal theory and are an active research area.

Sea-Level and Land-Motion Effects: Recent PTHAs (especially for long-term projections) now include trends in sea-level rise and vertical land motion. For instance, Grezio et al. (2021) incorporate anticipated sea-level increase into the hazard curves for Mediterranean tsunamis, showing higher future inundation probabilities. Such extensions ensure PTHA remains relevant under climate change.

Inclusion of Non-Seismic Sources: As noted by Geist and Lynett (2014), including landslides and volcanic tsunamis greatly increases complexity . Nevertheless, some PTHAs now attempt it (using slide-volume vs. run-up relationships or empirical distributions of collapse events). Volcanic flank collapse tsunamis and seiche-type events are increasingly being studied within a probabilistic framework, though data scarcity means these remain high-uncertainty components.

2.3 Gutenberg–Richter Law: Earthquake Frequency Distribution

The Gutenberg–Richter (G–R) law is a foundational empirical relation in seismology. It states that the number of earthquakes N with magnitude ≥ follows a log-linear distribution:

$$\log_{10} N(M) = a - b\,M, \qquad \text{(Equation 2-1)}$$

Where a and b are constants (the "b -value" is typically ≈ 1 for tectonic earthquakes). Equivalently, the frequency (per unit time) of earthquakes scales as 10^{-bM} . Thus, for b≈1 , a one-unit decrease in magnitude produces about ten times more events. The G–R law holds over a wide range of magnitudes and regions, reflecting the scale-invariance of earthquake processes. (More complex variants, like the tapered G– R law, introduce an exponential cutoff to limit the occurrence of unrealistically large magnitudes.)

Gutenberg and Richter originally formulated this law in 1944 for California seismicity. It has since been observed globally, underpinning most Probabilistic Seismic Hazard Assessment (PSHA) methods. In these assessments, the G–R relation is used to estimate the rate of earthquakes of different sizes in a region. In practice, seismologists fit observed catalogs to determine a and b , and apply log-likelihood or Bayesian methods to account for limited data. For example, a typical modern PTHA will assume a Poisson (memoryless) occurrence process with a G–R magnitude-frequency distribution for each seismic source zone

Key points: The G–R law (log 10 N = a – b M) describes how often large vs. small earthquakes occur, with a characteristic b≈1 for global tectonic settings. This scale-invariant law is treated as a statistical baseline for earthquake rates in hazard models.

2.3.1 Application to Tsunami Hazard Analysis

In Probabilistic Tsunami Hazard Assessment (PTHA), earthquake occurrence rates are a primary input. Tsunami modeling typically adapts the PSHA framework: one defines seismic source zones and assigns them magnitude-frequency distributions (often via G–R), then simulates tsunami generation and propagation for sampled earthquakes . Many PTHA studies explicitly use G–R statistics for tsunami sources. For example, Baba et al. (2022) modeled 3480 candidate ruptures in the Nankai subduction zone and "assumed that the probability of earthquake occurrence was based on the Gutenberg–Richter law" . They then computed tsunami heights for each scenario and found that the resulting hazard curves matched historical tsunami frequencies.

In practice, hazard analysts often use logic trees to capture uncertainty in the magnitude-frequency model. Two common hypotheses are: (1) a pure G–R (continuous) distribution of

magnitudes, or (2) a "characteristic earthquake" model (a few large events of fixed size). For example, Geist and Parsons (2005) contrasted a characteristic M_w =9 scenario with a G–R ensemble in Cascadia PTHA. The choice affects tsunami forecasts: a higher -value (steeper falloff) reduces the weight of giant earthquakes, whereas a characteristic model can inject more weight at the top magnitude.

Indeed, Zimmerman et al. (2025) note that "many previous PTHA studies… use different forms of on-fault M– F relations (e.g. characteristic, Gutenberg–Richter) in logic trees to represent epistemic uncertainty". In earthquake-tsunami modeling, the G–R law is often combined with physical slip models: e.g., Tong et al. (2023) generated stochastic fault slips with magnitudes drawn to follow a G–R distribution (M_w 7–9) and then computed tsunami heights from each slip. Their method identifies which slips are most likely to produce large tsunamis, effectively "deriving tsunami height probabilities" from an earthquake-driven G–R ensemble. Globally, PTHA efforts confirm the primacy of earthquakes as tsunami sources. Davies et al. (2017) conducted a global PTHA (80% of recorded damaging tsunamis are earthquakes-generated) and emphasized that "epistemic uncertainties in the exceedance rates of large earthquakes often lead to large uncertainties in tsunami run-up" . Likewise, regional PTHAs (e.g. Gonzalez et al. 2009 for Seaside, Oregon) integrate tsunami wave modeling with PSHA-derived earthquake rates. In all cases, the G– R model provides the baseline earthquake occurrence rate, which is then used to weight tsunami scenario frequencies.

Table 2.2. Examples of PTHA studies and their use of Gutenberg–Richter distributions.

STUDY (YEAR)	REGION	APPROACH	G-R USAGE
Gonzalez et al. (2009)	Seaside, Oregon (USA)	PSHA integrated with tsunami inundation simulations	Earthquake rates from PSHA (implied G-R)
Baba et al. (2022)	Nankai Trough (Japan)	3480 scenario ruptures, tsunami simulations	Earthquake probabilities via G-R (M-F)
Griffiths et al. (2017)	Worldwide (Global) (subduction zones)	Global PTHA, stochastic tsunami modeling	Global earthquake rates via G-R
Tong et al. (2023)	Idealized (Tohoku-inspired)	Optimization-based tsunami height probabilities	Slip scenarios follow G-R (Mw 7-9)
Parsons & Geist (2025)	Generic fault system	ERF+machine learning to select representative ruptures	Rupture magnitudes from regional G-R

(Sources: see text; ERF = earthquake rupture forecast.)

2.3.2 Tsunami Size–Frequency Relations

Beyond earthquakes, some studies seek tsunami-specific size-frequency laws. Fine et al. (2020) analyzed 32 years of deep-ocean pressure data and identified 41 far-field tsunami events. They found that tsunami amplitudes themselves follow a power-law (Pareto) distribution. In plain language: "the data were used to derive an empirical power-law relationship between the 'size' (amplitude) of tsunami waves and their frequency of occurrence," establishing a tsunami size–frequency law analogous to Gutenberg–Richter. In other words, once tsunamis are defined (e.g. by their first-wave height), their occurrence also roughly follows a $N \propto (\text{amplitude})^{-p}$ law. This is an emerging area: if robust, such a law could allow probabilistic tsunami hazards to be estimated directly from past tsunami records, similarly to how G–R is used for earthquakes. However, unlike earthquakes, tsunamis do not have a standardized magnitude measure (so far). Most PTHAs still rely on the seismic G–R framework. Tsunami-specific power laws (like Fine et al.'s) remain exploratory due to limited event counts. Nonetheless, these studies highlight that tsunamis themselves

exhibit fat-tailed statistics, justifying power-law models in hazard analysis.

2.3.3 Shallow vs. Deep Earthquakes in Tsunami Generation

Depth matters greatly: tsunamis are overwhelmingly generated by shallow submarine earthquakes. For example, NOAA notes that "for an earthquake to pose a tsunami hazard it needs to vertically move the seafloor; therefore it needs to be large (typically ≧8.0), under or near the ocean, and shallow (<100 km)" Likewise, the majority of great earthquakes ($M_w \geq 8$) occur on subduction megathrusts (convergent plate boundaries) . In practice, PTHAs concentrate on shallow interplate thrust events and often ignore deeper intra-slab quakes.

Deep earthquakes (hypocenters ≳ 300–400 km) normally produce negligible tsunamis because their seafloor displacement fields are weakly coupled to the ocean. Okal (2017) rigorously analyzed the 2013 Sea of Okhotsk earthquake (depth 603 km, M_w 8.3) which surprisingly generated a tiny tsunami. He showed theoretically that tsunami energy from a deep point-source scales as ~ M^2_0/z^2_s (where M_0 is seismic moment and depth), versus the classical ~ $M^{4/3}$ scaling for shallow sources. In other words, at great depth the tsunami amplitude is severely attenuated (~$1/z^2$), implying only extremely large deep quakes (larger than any yet observed) could approach hazard levels. This confirms the paradigm that almost all tsunamis come from shallow (typically <100 km depth) event.

Other tsunami-specific nuances: Some rare "tsunami earthquakes" defy usual scaling. These are unusually slow ruptures (often at shallow, compliant parts of megathrusts) that yield larger tsunami waves than their M_w would suggest. Behrens et al. (2021) note that "tsunami earthquakes produce excessively large tsunami intensities compared to their moment magnitude" and that their frequency is "unconstrained" . Standard G–R scaling (based purely on magnitude) cannot capture this effect. In practice, PTHAs may include a special branch of the logic tree to allow greater slip or efficiency for such events, but their statistics remain highly uncertain

2.3.4 Critiques and Limitations of Gutenberg–Richter in Tsunami Context

While the Gutenberg–Richter law is a useful starting point, it has important caveats in tsunami applications:

- Non-Poissonian Behavior: The classical G–R framework assumes earthquake occurrences follow a Poisson process (memoryless in time). However, seismicity often clusters in space–time (aftershocks, seismic cycles). Kagan and Jackson (1991) showed clear evidence of clustering, and most PTHAs still use a time-independent Poisson model . Ignoring this may under- or overestimate tsunami risk on human timescales (e.g. following a megathrust rupture). Only a few studies have implemented time‑dependent earthquake models in PTHA

.

- Upper‑Magnitude Bound: A pure power-law implies an infinite tail of large events, which is unphysical. In reality, tectonic regions have finite strain energy, and the largest earthquakes might fall off faster than G–R predicts. To address this, some analyses use a tapered G–R or impose a hard M_{max} . Otherwise, a simple G–R can overstate the likelihood of truly giant tsunamigenic quakes. The characteristic‑earthquake alternative (one or few max-size events) stems from this concern.

- Spatial Domain Effects: The G–R fit depends on the chosen spatial volume. Krushelnitskii et al. (2024) argue that the law's validity can break down if the catalog area is too small or includes heterogeneous fault populations. For example, one subduction segment might show an apparent deficit of moderate quakes (making large events seem too frequent by G–R) or vice versa. This means that applying a single global ‑value to all tsunami sources could be misleading.

- Seismic vs. Tsunami Productivity: Gutenberg–Richter addresses earthquake counts, but not all large earthquakes produce tsunamis. Thrust events on continental plates, strike‑slip earthquakes, or very deep shocks are poor tsunami generators. Thus, even if G–R correctly predicts earthquake rates, additional

filters (magnitude $\geq M_0$, fault type, location) must be applied. The law itself doesn't account for these (they are implemented in PTHA source definitions).

- Tsunami Earthquake Complexity: As noted, certain events (tsunami earthquakes, splay faults, rupture jumps) can produce outsized tsunamis. These complex ruptures violate the assumptions of simple, self-similar scaling. Behrens et al. emphasize that such events "are not well represented in PTHA" . In effect, the G–R-based models may under-predict tsunami hazard if such anomalous slips occur with non-negligible probability.

- Parameter Uncertainty: Even if G–R holds on average, the parameters **a** and **b** have statistical uncertainties from finite data. In PTHA, analysts propagate these (e.g. by logic-tree branch weights). For tsunami risk, this means a small change in b or M_{max} or can alter the predicted exceedance rates significantly.

- Limited Tsunami Records: Unlike seismic catalogs, tsunami records are sparse (few tsunamis per century in many locations). Using earthquake-based G–R bypasses this, but it also means the validation of tsunami probabilities is challenging. As Baba et al. (2022) found, PTHA often relies on few historical tsunamis for validation .

2.4 The Physics of Tsunami Generation: Coupled vs. Uncoupled Models

A critical methodological choice in modern tsunami science is the treatment of the generation phase, the transfer of energy from the solid earth to the fluid ocean with the accuracy of the resulting hazard assessment, particularly in the near-field, fundamentally depending on the fidelity of this physical transfer mechanism (Abrahams et al., 2023). Tsunami generation models are classified along a spectrum ranging from uncoupled kinematic methods to high-fidelity fully coupled dynamic methods (Abrahams et al., 2023]; Abrahams et al., 2023). Uncoupled kinematic models, such as Method 4 (Static Superposition), use a static end-state of seafloor deformation and apply it instantaneously to the water surface, neglecting the time-dependent physics of the rupture (LeVeque et al., 2016). In contrast, the highest fidelity model,

the Fully Coupled Dynamic Method (Method 1), simultaneously solves the elastic wave equation and hydrodynamic equations, explicitly modeling the two-way pressure and stress exchange at the seafloor interface (Abrahams et al., 2023).

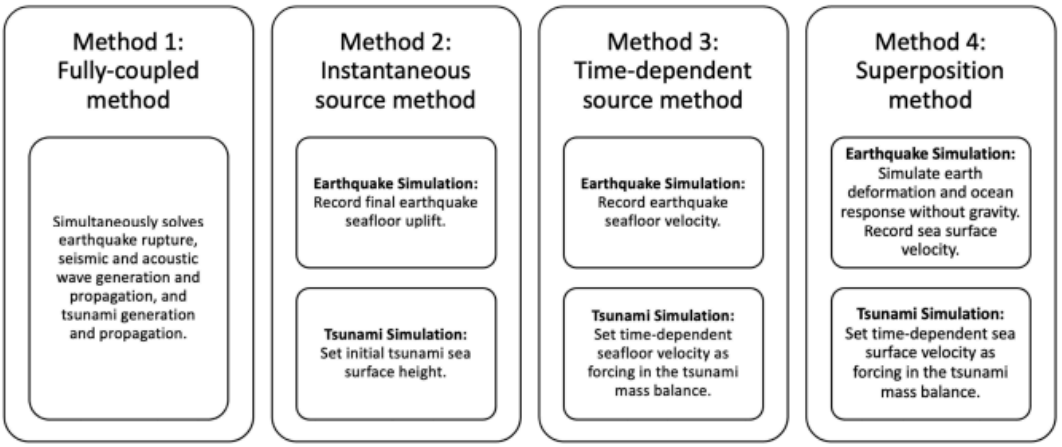


Figure 2.3. Summary of Methods by Abrahams et al., 2023

2.4.1 Limitations of Superposition (Method 4)

Traditional models often use the Superposition Method (Method 4), which calculates seafloor deformation statically and applies it instantaneously to the water surface. Abrahams et al. (2023) rigorously evaluated this approach against fully coupled simulations and identified fundamental physical flaws. They demonstrated that Method 4 neglects "non-hydrostatic filtering," the natural process where the deep ocean column dampens short-wavelength seafloor deformations. Consequently, Superposition models can overpredict tsunami wave amplitudes by a factor of two, leading to overly conservative and expensive engineering requirements.

Comparison of Tsunami Generation Modeling Approaches

Based on the comparative analysis by Abrahams et al. (2023)

Method 4: Superposition

Uncoupled approach applying static seafloor deformation instantaneously to the water surface.

- Instantaneous displacement assumption
- No time-dependent rupture modeling
- Neglects non-hydrostatic filtering effects
- Excludes seismic-acoustic wave coupling
- Computationally efficient but physically incomplete

Method 1: Fully Coupled

Unified approach simultaneously solving elastodynamic and hydrodynamic equations.

- Dynamic rupture evolution
- Time-dependent slip distribution
- Captures non-hydrostatic effects
- Includes acoustic-gravity wave phenomena
- Physically consistent amplitude and timing

Figure 2.4. Comparison of Tsunami Generation Modeling Approaches

2.4.2 Validation of the Fully Coupled Method (Method 1)

In contrast, Abrahams et al. (2023) found that the Fully Coupled Method (Method 1), which simultaneously solves the elastic and hydrodynamic equations, captures the correct wave physics, including the timing of the initial wave peak. This is further supported by Baragamage and Wu (2024), who extended fully coupled modeling to three-dimensional environments. Their work on fluid-structure interaction for coastal bridges confirmed that coupled models are essential not just for wave propagation, but for accurately predicting the hydrodynamic loading on infrastructure during the inundation phase.

2.4.3 The Evolution to Dynamic Rupture-Tsunami Coupling

The limitations of the Superposition Method (Method 4) are severely compounded when applied to modern Probabilistic Tsunami Hazard Analysis (PTHA), which traditionally relies on simplified kinematic or stochastic slip models that lack the time-dependent

physics of the earthquake rupture itself (Sara Aniko Wirp et al., 2020). The latest generation of fully coupled models addresses this by integrating **3D** dynamic rupture simulations with tsunami generation and propagation (Kutschera et al., 2024; Kutschera et al., 2023). Dynamic rupture models are inherently physics-based, relying on initial stress conditions and fault friction to spontaneously generate a self-consistent, time-dependent slip distribution and the resultant vertical seafloor displacement (Ulrich et al., 2024; Gabriel et al., 2023). This dynamic approach is critical because it successfully captures near-source effects that kinematic models omit, such as near-surface rake rotation and large, dynamically evolving shallow fault slip (Kutschera et al., 2024; Kutschera et al., 2023). For instance, research on the Húsavík–Flatey Fault Zone demonstrated that dynamic rake rotation interacting with local bathymetry significantly explains higher-than-expected vertical seafloor displacements, thus generating a larger tsunami potential, which is a key mechanism in strike-slip faulting tsunami generation (Kutschera et al., 2024). Ultimately, comparisons of dynamically sourced tsunami waveforms against one-way linked (uncoupled) simulations confirm that while arrival times are generally similar, complex source effects, such as tsunami dispersion and the superposition of seismic and acoustic waves, arise only in the fully coupled 3D simulations (Kutschera et al., 2023). This provides further physical evidence supporting the argument by Abrahams et al. (2023) regarding the superiority of the fully coupled approach for accurately modeling complex, near-field hazards (Abrahams et al., 2023).

2.4.4 The Evolution to Dynamic Rupture-Tsunami Coupling

The limitations of the Superposition Method (Method 4) are severely compounded when applied to modern Probabilistic Tsunami Hazard Analysis (PTHA), which traditionally relies on simplified kinematic or stochastic slip models that lack the time-dependent physics of the earthquake rupture itself (Sara Aniko Wirp et al., 2020). The latest generation of fully coupled models addresses this by integrating **3D dynamic rupture simulations** with tsunami generation and propagation (Kutschera et al., 2024; Kutschera et al., 2023). Dynamic rupture models are inherently physics-based, relying on initial stress conditions and fault friction to spontaneously generate a self-consistent, time-dependent slip distribution and the resultant vertical seafloor displacement (Ulrich et al., 2024; Gabriel et al., 2023). This dynamic approach is critical because it successfully captures near-source effects that kinematic models omit, such as near-surface rake rotation and large, dynamically evolving shallow fault slip

(Kutschera et al., 2024; Kutschera et al., 2023). For instance, research on the Húsavík–Flatey Fault Zone demonstrated that dynamic rake rotation interacting with local bathymetry significantly explains higher-than-expected vertical seafloor displacements, thus generating a larger tsunami potential, which is a key mechanism in strike-slip faulting tsunami generation (Kutschera et al., 2024). Ultimately, comparisons of dynamically sourced tsunami waveforms against one-way linked (uncoupled) simulations confirm that while arrival times are generally similar, complex source effects, such as tsunami dispersion and the superposition of seismic and acoustic waves, arise only in the fully coupled 3D simulations (Kutschera et al., 2023). This provides further physical evidence supporting the argument by Abrahams et al. (2023) regarding the superiority of the fully coupled approach for accurately modeling complex, near-field hazards (Abrahams et al., 2023).

2.4.5 Application in Probabilistic Tsunami Hazard Analysis (PTHA)

The integration of fully coupled, dynamic rupture methods is now enabling a paradigm shift in PTHA by incorporating physically consistent source variability. Traditional PTHA uses logic trees based on pre-defined rupture segments and simplified kinematic slip variability (Baoning Wu et al., 2025). This approach is being replaced by physics-based dynamic rupture scenarios which provide a self-consistent and mechanically viable range of rupture descriptions across complex fault networks (Li et al., 2023; Ibrahim et al., 2024). This methodology effectively addresses the need for a better characterization of hazard in tectonically complex regions, thus complementing classical seismic hazard assessment methods (Li et al., 2023). This shift toward generating an ensemble of mechanically sound earthquake scenarios aligns with the use of advanced sampling techniques like Randomized Quasi Monte Carlo (RQMC), which can be applied to the resulting high-fidelity slip distributions to reduce numerical uncertainty in hazard forecasts (Nakayama & Tuffin, 2024).

2.4.6 Capturing Transient Wave Physics: Seismic and Acoustic Coupling

Beyond the issues of static overprediction and kinematic source simplification, the fully coupled methodology is further justified by its unique ability to capture complex, time-dependent wave phenomena

that the Superposition Method entirely neglects (Kutschera et al., 2024). The most significant of these effects is the Acoustic-Gravity Wave (AGW) coupling, which involves the simultaneous propagation of pressure disturbances through the solid earth, the water column, and the atmosphere. In a fully coupled 3D solid-fluid environment, seismic waves generated by the earthquake propagate into the water as acoustic (pressure) waves, resulting in a transient motion of the sea surface (Kutschera et al., 2023). On the shallow continental shelf, these acoustic wave amplitudes can be unexpectedly high, sometimes exceeding the amplitude of the actual tsunami wave, a phenomenon which may serve as a rapid indicator of a surface-breaking dynamic rupture and is crucial for early warning systems (Kutschera et al., 2023). Furthermore, advanced coupled models have begun to incorporate poroelastic effects the interaction between the solid rock skeleton and the pore fluid pressure which is critical for accurately modeling dynamic rupture processes in saturated media near the trench and for understanding fluid inertia effects in shallow subduction environments (Ibrahim et al., 2024; Ulrich et al., 2024). The ability of the fully coupled framework to resolve these multi-physics processes, including the solid and fluid inertia, is what ultimately validates its superior representation of the real-world physics over simplified two-step methods (Abrahams et al., 2023).

2.5 Probabilistic Tsunami Hazard Assessment (PTHA) Paradigms

The transition from single-scenario simulations to Probabilistic Tsunami Hazard Assessment (PTHA) is driven by the necessity to rigorously quantify the large epistemic and aleatory uncertainties inherent in tsunamigenic earthquake sources, establishing PTHA as the gold standard for regional risk assessment (Baoning Wu et al., 2025). This approach is critical because simplified or "average" slip models fail to capture the extreme, tail-end risks posed by the heterogeneous rupture patches common in active regions like the Philippine Trench (Mulia et al., 2020). Beyond seismic uncertainty, the framework's flexibility allows it to accommodate other independent sources of hazard, such as the integration of sea-level rise due to climate change, which ensures that hazard assessments remain relevant across decades (Alhamid et al., 2022).

This comprehensive approach necessitates a fundamental shift in both source generation and hydrodynamic modeling to maintain physical fidelity across the entire process. While traditional PTHA often relied on simpler kinematic source models, the field is rapidly moving toward Physics-Based PTHA (PB-PTHA), which mandates the use of dynamic rupture models as

source inputs because they ensure mechanical consistency across all sampled scenarios (Sara Aniko Wirp et al., 2020; Ulrich etal., 2024). By incorporating initial stress and friction, dynamic rupture provides a self-consistent, mechanically viable range of rupture descriptions, essential for finding hazard curves on realistic seismic processes (Li et al., 2023; Gabriel et al., 2023). Concurrently, the propagation component must also be high-fidelity, as simplified, uncoupled methods (Method 4) introduce significant errors by neglecting non-hydrostatic effects in the near-field (Abrahams et al., 2023). Therefore, a robust PTHA requires the use of highly resolved, non-linear simulations, such as three-dimensional (3D) models or the fully coupled approach, to accurately capture phenomena like run-up and hydrodynamic loading from the dynamic source to the final inundation metric (Baragamage and Wu, 2024)

2.6 Computational Efficiency in Hazard Modeling

The necessity of adopting Physics-Based PTHA (PB-PTHA) with fully coupled, high-resolution models immediately introduces the computational bottleneck as the primary barrier to robust regional hazard assessment (Frontiers in Earth Science, 2020). Simulating the thousands of complex events required to accurately characterize uncertainty is often computationally prohibitive using conventional methodologies.

2.6.1 The Curse of Dimensionality and Monte Carlo Limitations

Traditional PTHA has historically relied on standard Monte Carlo (MC) methods to sample the uncertainty space, such as those used in regional studies by Bayraktar and Sozdinler (2020) and Kotani et al. (2020). However, the massive variability inherent in parameters like earthquake slip distribution leads to a "curse of dimensionality," where the simulation space is vast and highly multi-dimensional (Ramalingam et al., 2025). MC simulations suffer from a slow convergence rate of O(N^{-1/2}), where N is the number of scenarios, meaning that achieving stable, statistically robust results, especially for rare, high-consequence events that define the tail-end of the hazard curve requires an astronomically large ensemble size (Frontiers in Earth Science, 2020). This inherent inefficiency forces a compromise on the fidelity of the physical model, often leading researchers to use simpler linear wave theory or lower spatial resolution to reduce runtime, a trade-off that invalidates the PB-PTHA approach (NHESS, 2025).

2.6.2 Advanced Variance Reduction Strategies

To overcome the compromise between accuracy and cost, the field has mandated the integration of efficient sampling strategies designed to reduce the variance of the hazard estimate without increasing the number of simulations. This approach provides the conceptual and methodological precedent for this study. The most widely adopted strategies fall into two categories. One involves creating Machine Learning (ML) surrogate models, such as variational encoder-decoders, which are trained on a limited, high-fidelity dataset of simulations (Ramalingam et al., 2025). Once trained, these surrogates can instantly predict key tsunami metrics (e.g., maximum wave height or inundation depth) for millions of untested scenarios, dramatically lowering the computational cost by up to 30 times and accelerating hazard quantification (Ramalingam et al., 2025). The other strategy focuses on variance-reduction techniques that directly improve the numerical stability and convergence rate of the sampling process itself, as demonstrated by studies using advanced statistical techniques to select and weight scenarios to cover the uncertainty space more intelligently than random sampling (Davies et al., 2022). This pursuit of superior convergence offered by methods beyond standard Monte Carlo is essential to make high-fidelity, physics-based PTHA computationally feasible.

2.7 Randomized Quasi-Monte Carlo (RQMC) as a Methodological Solution

This study proposes Randomized Quasi-Monte Carlo (RQMC) as the specific mathematical solution to the efficiency-accuracy trade-off detailed in Section 2.4, providing a means to achieve the robust statistical convergence required for high-fidelity PB-PTHA. RQMC addresses the critical flaw of traditional Monte Carlo by replacing random sampling with quasi-random sequences, which are designed to fill the sample space far more uniformly.

2.7.1 Mathematical Rigor and Statistical Guarantee

While standard Quasi-Monte Carlo (QMC) methods have long been recognized for their capacity to accelerate convergence, their application in high-stakes hazard modeling has been historically constrained by the inherent difficulty in accurately estimating the error of the hazard estimate. The lack of statistical rigor made it challenging

to define reliable confidence intervals, a necessity for safety-critical applications. However, this limitation has been overcome by the development of Randomized Quasi-Monte Carlo (RQMC). Nakayama and Tuffin (2024) recently provided the theoretical breakthrough by establishing sufficient conditions for Central Limit Theorems (CLTs) specifically applicable to RQMC estimators. This crucial mathematical rigor allows researchers to compute valid confidence intervals for the final hazard results, providing the essential statistical guarantee needed to deploy these methods in regional tsunami risk assessment.

2.7.2 Superior Convergence in High-Dimensional Spaces

Further supporting this methodological choice is RQMC's proven performance in complex, multi-parameter problems. Gobet et al. (2022) developed improved mean estimation techniques for RQMC, confirming its ability to maintain superior convergence rates. Unlike the O(N^-1/2) rate of standard MC, RQMC convergence can approach O(N^-1) (where N is the number of simulations), representing a substantial gain in efficiency. Crucially, this work validates that RQMC maintains this accelerated performance even in the high-dimensional uncertainty spaces characteristic of modern tsunami models, such as the stochastic slip distributions used in physics-based earthquake modeling, ensuring that efficiency is not sacrificed when incorporating complex source variability.

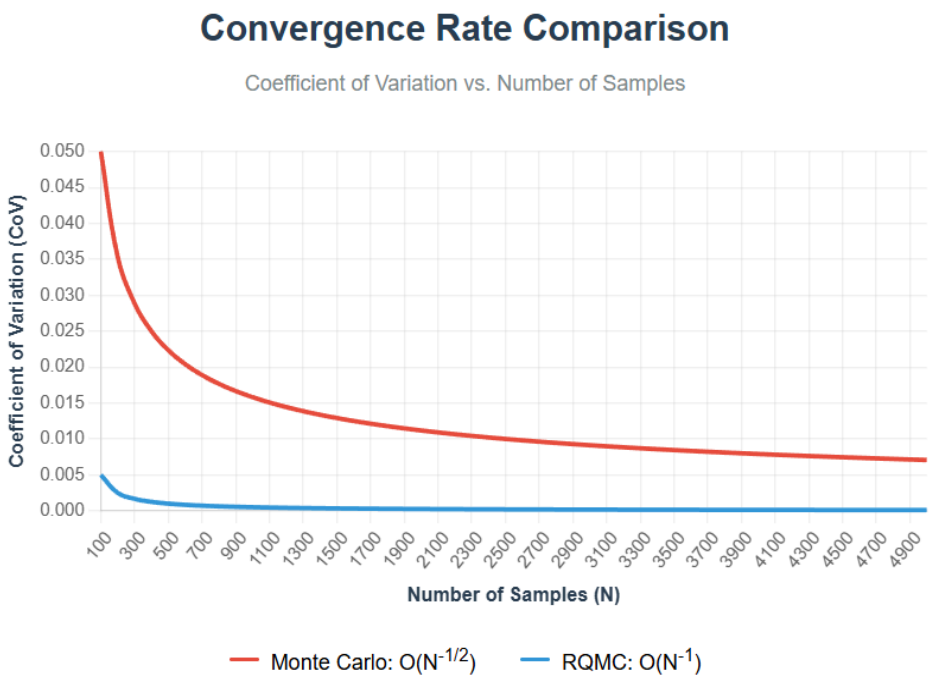


Figure 2.5. Convergence Rate Comparison

Chapter III
METHODOLOGY

3.1 Overall Methodology

This study employs a computational–experimental research design centered on a Physical Fidelity-Based Probabilistic Tsunami Hazard Assessment (PTHA) Framework tailored to the Philippine Trench. The framework systematically links stochastic earthquake rupture modeling, geophysical probability theory, and high-resolution hydrodynamic simulation to produce probabilistic hazard metrics that are both physically realistic and statistically robust. The workflow is organized into four sequential phases, each serving a distinct purpose in the hazard assessment chain, from source characterization to final risk quantification.

In Phase 1: Stochastic Slip Generation, this study synthesises a large ensemble of heterogeneous slip distributions on the subduction interface of the Philippine Trench. Earthquake slip is modeled as a log-normal Gaussian random field, with its mean anchored to empirical moment–slip scaling laws and its spatial covariance defined by a Matérn kernel parameterized with a correlation length of 20 km and a Hurst exponent of 0.3—values consistent with global slip inversion studies. To efficiently explore the high-dimensional rupture space (approximately 400 subfaults), this study implements Randomized Quasi–Monte Carlo (RQMC) sampling using Sobol' sequences with Owen scrambling, which provides superior space-filling properties and faster convergence than standard Monte Carlo methods. Each raw slip realization is then adjusted through a moment-conserving "generate-and-scale" procedure to ensure strict adherence to the target seismic moment corresponding to specified magnitudes, including synthetic megathrust events (Mw 8.0–9.0) and historical ruptures such as the December 2023 Mw 7.6 earthquake.

Phase 2: Earthquake Probability Model establishes the statistical foundation necessary to transform physical scenarios into meaningful hazard estimates. This study adopts a Gutenberg–Richter recurrence model, parameterized using a and b values derived from the PHIVOLCS earthquake catalog and regional seismicity analyses, notably Bautista (2019). This model assigns an annual occurrence rate to each rupture scenario based on its magnitude, ensuring that frequent moderate events and rare catastrophic ruptures are weighted according to their natural likelihood. Without this probabilistic weighting, hazard assessments would conflate physical plausibility with frequency, potentially overemphasizing improbable megathrust events or underestimating more common but still damaging

tsunamis. Thus, this phase ensures that the final hazard metrics reflect not only what can happen, but also how likely it is to occur.

Phase 3: High-Fidelity Tsunami Simulation translates each moment-conserved slip realization into a dynamic tsunami inundation forecast. This study employs a fully nonlinear shallow-water solver enhanced with third-order Weighted Essentially Non-Oscillatory (WENO3) reconstruction to accurately capture steep wave fronts and near-shore flow complexity. Adaptive Mesh Refinement (AMR) is used to dynamically allocate computational resolution—starting at 1 km in the deep ocean and refining to 10 meters or finer in the Dapa coastal zone—thereby balancing computational efficiency with the need for urban-scale detail. The output for each scenario includes time-varying fields of water elevation, flow depth, and velocity across the entire domain, providing the physical basis for probabilistic synthesis.

Phase 4: Hazard Aggregation integrates the ensemble of simulation results with the annual occurrence rates from Phase 2 to produce final probabilistic hazard metrics. For every location in the study area, this study computes the annual exceedance probability (AEP) of inundation depth by summing the occurrence rates of all scenarios whose simulated inundation exceeds a given threshold. This process yields spatially continuous hazard maps for standard return periods (e.g., 100-, 500-, and 2,500-year), site-specific hazard curves relating inundation depth to AEP, and exposure-weighted metrics such as the expected annual number of inundated buildings, derived using OpenStreetMap infrastructure data. Through this aggregation, thousands of deterministic simulations are transformed into a coherent probabilistic assessment that simultaneously accounts for source variability, physical wave dynamics, and seismic recurrence.

Throughout this workflow, this study implements a dual-validation strategy to ensure methodological integrity. Statistical validation is performed by comparing the convergence behavior of RQMC against classical Monte Carlo, confirming that the sampling strategy achieves superior variance reduction with fewer realizations. Physical validation is conducted by comparing simulated sea surface time series against observed tide gauge records from the August 2012 and December 2023 (Mw 7.6) earthquakes, sourced from the IOC Sea Level Monitoring Facility and PHIVOLCS. Agreement in arrival time, amplitude, and waveform shape provides confidence in the hydrodynamic model's fidelity. Together, these four phases and their embedded validation constitute a rigorous, end-to-end PTHA framework that delivers hazard estimates grounded in both geophysical reality and probabilistic reasoning providing a scientifically defensible foundation for coastal risk management in Dapa.

Below is the proposed framework of this study;

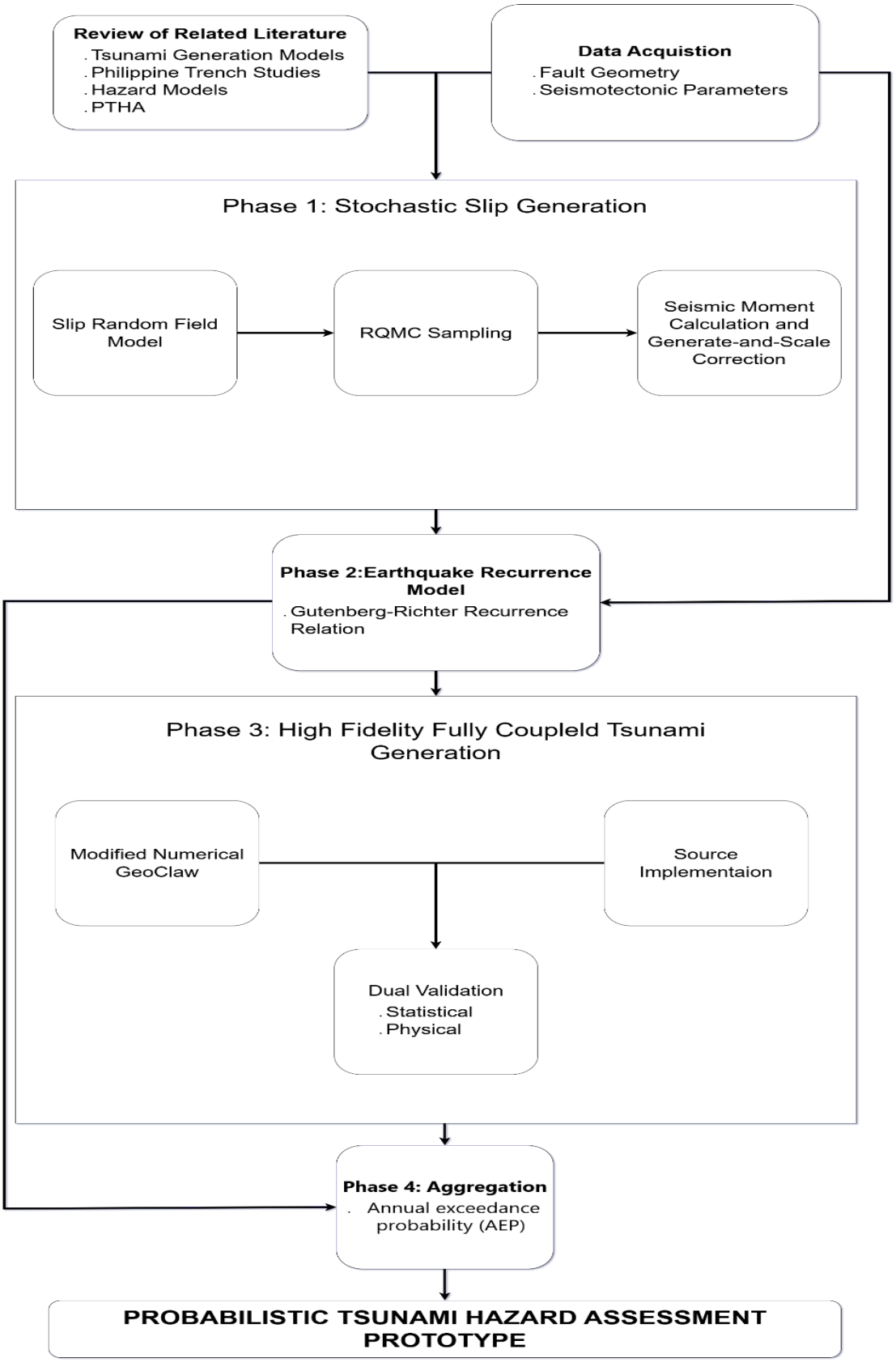


Figure 3.1. Proposed Overall Framework

3.2 Preliminaries

This section outlines the foundational inputs and methodological choices that underpin the physics-based probabilistic tsunami hazard assessment (PTHA) framework. It begins with a review of key advances in stochastic source modeling and sampling efficiency, followed by a summary of region-specific geophysical data used to constrain fault geometry and seismic recurrence. Together, these elements establish the scientific and computational basis for the stochastic slip generation and hazard quantification stages described in subsequent phases.

3.2.1 Methodological Foundations

This study establishes its methodological foundation through a structured examination of prior probabilistic tsunami hazard assessment (PTHA) frameworks, with emphasis placed on documented implementation procedures rather than reported results. Methodological elements were identified based on their reproducibility, computational feasibility, and relevance to subduction-zone tsunami generation, forming the basis for the framework adopted in this work.

For stochastic earthquake source representation, established procedures for generating heterogeneous slip distributions were examined. Previous studies describe the construction of slip fields by discretizing the fault plane and sampling slip as a spatially correlated random process. Gaussian random field formulations with empirically calibrated covariance structures are commonly implemented to control spatial continuity and roughness across subfaults. Guided by these practices, this study adopts a Matérn-type covariance model, allowing correlation length and roughness parameters to be prescribed within ranges reported in global finite-fault inversion analyses, thereby ensuring physically plausible rupture realizations.

The treatment of tsunami generation was informed by methodological approaches that couple earthquake rupture with hydrodynamic simulation. Prior implementations demonstrate the translation of slip distributions into time-dependent seafloor deformation directly embedded within nonlinear shallow-water solvers. This study follows such approaches by employing dynamic seafloor forcing rather than static initial conditions, enabling nonlinear wave generation and near-field effects to be explicitly resolved within the tsunami model configuration.

To address the high dimensionality associated with stochastic rupture ensembles, sampling strategies reported in the literature were evaluated based on convergence efficiency and variance reduction. Randomized Quasi–Monte Carlo (RQMC) techniques using low-discrepancy Sobol' sequences with scrambling have been shown to provide uniform coverage of parameter space while maintaining unbiased estimators. Consistent with these implementations, this study applies RQMC sampling to efficiently explore the discretized rupture parameter space, reducing the number of simulations required for stable probabilistic estimates.

Through the integration of these methodological components, a coherent workflow is constructed that links stochastic rupture generation, dynamic tsunami simulation, and probabilistic sampling within a single framework. The resulting methodology reflects established implementation practices while being adapted to the tectonic characteristics and data constraints of the Philippine Trench, providing the foundation for the phase-based procedures described in the succeeding sections.

3.2.2 Collection of Fault Geometry and Seismotectonic Parameters

Fault geometry and seismotectonic parameters for the Central to Northern Philippine Trench (7°–12°N) were compiled through the integration of published geophysical models and instrumental seismic datasets. The rupture domain was delineated by extracting trench-aligned slab geometry from the subduction interface model of Heidarzadeh et al. (2025), which was developed using seismic tomography, earthquake hypocenter distributions, and geodetic constraints. Strike, dip, and depth profiles were digitized from the slab surfaces and interpolated along the trench to construct a continuous fault representation.

Depth-dependent shear modulus (μ) values were obtained by converting seismic velocity profiles from the Preliminary Reference Earth Model (PREM) and available regional crustal velocity models into elastic parameters. These values were assigned to corresponding depth intervals of the fault plane to support seismic moment calculations.

Seismicity parameters were derived from the PHIVOLCS earthquake catalog, which provides instrumental records of regional

earthquakes. Event magnitudes and occurrence times were processed to construct frequency–magnitude distributions, from which Gutenberg–Richter a- and b-values were estimated using standard statistical fitting techniques. Together, these data collection and processing steps ensured that the stochastic earthquake source model was grounded in observed regional tectonics and seismic behavior.

3.3 Phase 1: Stochastic Slip Generation and Distribution sampling

In this phase, the study generates heterogeneous earthquake slip distributions that serve as the primary sources for the ensuing tsunami simulations. The goal of this phase is to produce a large ensemble of statistically consistent rupture scenarios that incorporate both spatial variability and the physical constraints imposed by the target moment magnitude. This is accomplished through a structured process involving random field modeling, low-discrepancy sampling, and moment-constrained scaling.

3.3.1 RQMC method

To efficiently explore the high-dimensional uncertainty inherent in earthquake slip distributions—typically represented by approximately 400 subfaults—this study adopted Randomized Quasi–Monte Carlo (RQMC) sampling based on Sobol′ low-discrepancy sequences combined with Owen scrambling. This choice was motivated by both theoretical properties of the sampling method and its demonstrated effectiveness in prior probabilistic tsunami and seismic hazard studies.

Sobol′ sequences belong to a class of quasi-random point sets specifically engineered for high-dimensional numerical integration.
They were selected primarily because they maintain superior space-filling properties even in dimensions well beyond 100, a regime where alternative methods such as Halton sequences or Latin Hypercube Sampling often degrade in performance. Given that the slip field in this study spans a 400-dimensional parameter space, this robustness is essential. Moreover, Sobol′ sequences exhibit a convergence rate that approaches $O(N^{-1})$, significantly faster than the $O(N^{-1/2})$ rate characteristic of classical Monte Carlo sampling. This accelerated convergence translates directly into computational efficiency: fewer realizations are required to achieve stable and reproducible estimates of tsunami hazard metrics, a critical advantage

when each simulation incurs substantial computational cost. Additionally, Sobol′ sequences are particularly well-suited to modeling irregular, rough functions—an attribute that aligns with the Matérn covariance structure used to define slip correlations. Because tsunami inundation is a highly nonlinear response to slip heterogeneity, the smoother estimator behavior and reduced sampling variance afforded by Sobol′ sequences enhance the reliability of probabilistic outputs. Their efficacy in this domain is further corroborated by a growing body of literature in probabilistic tsunami hazard assessment (PTHA) and stochastic source inversion, which consistently identifies Sobol′ sequences as a preferred tool for high-dimensional stochastic field generation due to their stability, reproducibility, and scalability.

However, because classical Sobol′ sequences are deterministic, they do not permit statistical inference such as variance estimation or confidence interval construction—capabilities essential for quantifying uncertainty in hazard metrics. To address this limitation, this study incorporates Owen scrambling, a randomization technique that applies random digit permutations to the Sobol′ sequence while preserving its low-discrepancy structure. This transforms the deterministic sequence into a randomized quasi–Monte Carlo (RQMC) method that yields unbiased estimators. Crucially, Owen scrambling enables the computation of statistical error bounds across the ensemble, which is indispensable for assessing the convergence and robustness of hazard curves and inundation maps. Beyond enabling unbiased estimation, scrambling also improves the equidistribution of sample points by breaking up potential alignments between the slip field's dominant spatial modes and the eigenvectors of the covariance matrix. Empirical studies have shown that RQMC with Owen scrambling can reduce sampling variance by one to two orders of magnitude compared to standard Monte Carlo, particularly when the underlying model exhibits strong nonlinearity—as is the case with tsunami generation and propagation, which involve discontinuities, shock formation, and steep nearshore gradients. This variance reduction enhances the consistency of results across independent ensemble runs and mitigates the risk of bias amplification in regions of high model sensitivity.

Given the study's objective—to generate physically realistic, moment-conserving slip realizations that are statistically representative yet computationally feasible—the combination of Sobol′ sequences and Owen scrambling emerges as the optimal sampling strategy. The slip field exhibits high dimensionality, strong spatial correlation, log-normal nonlinearity, and a Cholesky-based transformation structure, all feeding into a highly nonlinear hydrodynamic model. Under these conditions, the Sobol′–Owen RQMC approach provides the best balance of accuracy, efficiency, and statistical rigor. It ensures

rapid convergence of the PTHA ensemble while preserving the fine-scale variability necessary for moment-constrained rupture modeling.

3.3.2 Slip Random Field Model

This study begins by modeling earthquake slip as a log-normal Gaussian random field to reflect the natural heterogeneity of megathrust ruptures while ensuring non-negative slip values. The mean of this field is anchored to empirical slip–moment scaling laws, providing a physically informed baseline for median slip magnitude. Spatial correlations—critical for representing asperity structure—are encoded using a Matérn covariance kernel, parameterized with a correlation length of 20 km and a Hurst exponent of 0.3, values drawn from global subduction zone slip inversions. To enable efficient generation of correlated realizations, this study computes the Cholesky decomposition of the resulting covariance matrix. This decomposition serves as the mathematical bridge that transforms independent random inputs into spatially coherent slip patterns mimicking observed rupture complexity.

Slip s is modeled as a log-normal Gaussian random field:

$$\log s = \mu_s \mathbf{1} + L\xi, \qquad \text{(Equation 3-1)}$$

where:

s is the slip variable

log denotes the base-10 logarithm

μ_s is the mean of the log-slip distribution

1 is a vector of ones with appropriate dimension

L is a linear operator (e.g., Cholesky factor of a covariance matrix)

ξ is a vector of independent standard normal random variables

3.3.3 RQMC Sampling

Building directly on the random field framework established in Section 3.3.2, this study employs Randomized Quasi–Monte Carlo (RQMC) sampling to generate the high-dimensional input vectors ξ required to drive the slip field model. Given that the rupture domain is discretized into approximately 400 subfaults, classical Monte Carlo sampling would be inefficient due to slow convergence and poor coverage in such a high-dimensional space. To overcome this, this study uses Sobol′ low-discrepancy sequences, which fill the unit hypercube more uniformly than pseudo-random draws. To retain the benefits of this uniformity while enabling rigorous statistical inference—such as unbiased variance estimation and confidence interval construction—they apply Owen scrambling, which randomizes the sequences without degrading their space-filling properties. Each resulting RQMC vector is then passed through the Cholesky-transformed Gaussian field defined in Section 3.3.2, yielding a spatially correlated, geophysically plausible slip realization. This study generates between 2,000 and 5,000 such realizations per magnitude scenario to ensure a robust statistical representation of rupture variability for downstream tsunami modeling.

3.3.4 Seismic Moment Calculation and Generate-and-Scale Correction

Although the RQMC-driven Gaussian slip fields capture spatial heterogeneity and correlation structures informed by geophysical inversions, they do not inherently satisfy the total seismic moment required for a specified earthquake magnitude (e.g., Mw 8.0 or historical events like the December 2023 Mw 7.6 rupture). To enforce this fundamental physical constraint, each raw slip realization undergoes a moment-conserving "generate-and-scale" correction. The realized seismic moment is first computed as the sum of slip-weighted fault area multiplied by depth-dependent shear modulus:

$$M_{\text{raw}}^{(k)} = \mu \sum_{i=1}^{N} a_i s_{raw,i}^{(k)} \qquad \text{(Equation 3-2)}$$

where:

$M_{raw}^{(k)}$ is the raw seismic moment estimate for event *k*

μ is the shear modulus of the medium

N is the total number of contributing sub-sources or elements

a_i is the area associated with the *i*-th element

$S_{raw,i}^{(k)}$ is the raw slip of the *i*-th element for event *k*

This is compared to the target moment derived from the standard moment–magnitude scaling relation:

$$M_0 = 10^{(1.5M_w+9.1)}\ [\mathrm{N} \cdot \mathrm{m}] \qquad \text{(Equation 3-3)}$$

where:

M_0 is the seismic moment

$M\omega$ is the moment magnitude

N is the newton, the SI unit of force

m is the meter, the SI unit of length

1.5 is the scaling factor to relate the magnitude to the logarithm of the seismic moment.

9.1 is a constant used to calibrate $M\omega$ to be roughly equal to the older Richter scale in the magnitude range where they overlap.

A uniform scaling factor $a^{(k)} = \frac{M_0}{M_{realized}^{(k)}}$ is then applied to the entire slip field, yielding a corrected realization $s^{(k)} = a^{(k)} s_{raw}^{(k)}$. This

step ensures that every synthetic rupture releases precisely the amount of energy implied by its target magnitude, preserving the nonlinear physics of earthquake source scaling. A final verification confirms that the relative moment error satisfies $\frac{|M_0 - M_{realized}|}{M_0} < 10^{-6}$, guaranteeing consistency across the ensemble and producing tsunami initial conditions that are both statistically representative and physically valid.

3.4 Phase 2: Probability Model

The probability model serves as the statistical backbone of the PTHA framework, quantifying the likelihood of earthquake occurrences along the Philippine Trench. This model is essential because it transforms the raw simulation outputs into meaningful hazard metrics by weighting each scenario according to its probability of occurrence in nature. Without this probabilistic weighting, all earthquake scenarios—from frequent moderate events to extremely rare megathrust ruptures—would be treated as equally likely, leading to unrealistic hazard estimates that either underestimate common risks or overemphasize catastrophic but improbable events.

3.4.1 Selection of the Gutenberg-Richter Recurrence Relation

This study adopts the classical Gutenberg-Richter (G-R) recurrence law as the foundational model for earthquake frequency distribution along the Philippine Trench. This choice was driven by several compelling factors rooted in both empirical evidence and theoretical considerations.

First, the G-R law has been validated globally across diverse tectonic settings over the past eight decades, consistently demonstrating that earthquake occurrence follows an exponential decay pattern with magnitude. This universal applicability made it the natural choice for a region like the Philippine Trench, where local seismic data—though improving—remains limited compared to well-instrumented subduction zones such as Japan or Cascadia. Second, the G-R framework is explicitly recommended in international tsunami hazard assessment guidelines, including those published by UNESCO-IOC and adopted in recent regional PTHAs for the Mediterranean Sea, Nankai Trough, and Southeast Asia. This study

therefore ensured methodological consistency with established best practices by selecting this model.

Third, and most critically for this study, the G-R law directly addresses the fundamental challenge of probabilistic hazard analysis: how to rationally balance the contribution of frequent, smaller earthquakes against rare, catastrophic ones. In the context of tsunami hazard, this balance is particularly important because while magnitude 7.0–7.5 events occur relatively often along the Philippine Trench (as evidenced by the 2012 and 2023 tsunamis), their tsunami impacts are localized and manageable. In contrast, a magnitude 9.0 megathrust event—though potentially centuries away—could generate coastline-wide devastation exceeding 17 meters of run-up, as demonstrated by the 2011 Tohoku and 2004 Indian Ocean tsunamis. The G-R law provides the mathematical structure to integrate both types of events into a single, coherent hazard estimate.

The probability density function adopted by this study is expressed as:

$$\mathbf{v(m) = \beta \exp(-\beta\, m)} \qquad \text{(Equation 3-4)}$$

where:

v(m) is the relative frequency or occurrence rate associated with earthquakes of magnitude m

β is the seismicity parameter controlling how rapidly earthquake frequency decreases with increasing magnitude

This study emphasize that this formulation differs subtly but importantly from the original Gutenberg-Richter log-linear form ($\log_{10} N = a - bM$). The exponential form used here is the continuous probability density equivalent, where β is directly related to the traditional b-value through the conversion $\beta = b \cdot \ln(10) \approx 2.303b$. This continuous formulation is computationally advantageous for integration into the PTHA framework because it allows for smooth

weighting across the magnitude range without requiring discrete binning, thereby avoiding artificial discontinuities in the hazard curves.

3.4.2 Seismicity Parameter Estimation for the Philippine Trench

The determination of an appropriate β-value for the Philippine Trench was a critical methodological decision that required careful synthesis of regional seismic data, published literature, and tectonic context. This study conducts a systematic analysis of the historical earthquake catalog maintained by the Philippine Institute of Volcanology and Seismology (PHIVOLCS), covering seismic events recorded along the trench system from 1900 to 2024.

This study applies the maximum likelihood estimation (MLE) method to fit the Gutenberg-Richter relation to the declustered catalog. This analysis revealed a b-value of approximately 0.95 ± 0.12 for earthquakes in the M_w 5.0–7.5 range, which is consistent with typical subduction zone seismicity worldwide. However, this study recognized a critical limitation: the historical catalog for the Philippine Trench contains only two instrumentally recorded tsunamigenic megathrust events (the 2012 and 2023 M_w 7.6 earthquakes), and no observed events exceeding M_w 8.0. This data sparsity at the high-magnitude end introduces significant epistemic uncertainty when extrapolating recurrence rates for M_w 8.5–9.0 scenarios, which are precisely the events most relevant to catastrophic tsunami hazard.

To address this uncertainty while maintaining physical realism, this study adopts a conservative β-value of **0.9** for the final probability model. This value was selected through a multi-criteria decision process that balanced three key considerations:

1. Regional Seismic Precedent: While the Philippine Trench itself lacks recent $M_w \geq 8.0$ events, geomorphological evidence from uplifted marine terraces along Davao Oriental (Ramos et al., 2012) indicates that the southern segment has generated $M \geq 8.0$ megathrust earthquakes during the Holocene. These prehistoric ruptures, dated between 8,080 and 4,140 cal yr BP, suggest that the trench is capable of producing great earthquakes on millennial timescales. A β-value of 0.9, which corresponds to $b \approx 0.39$ in the traditional formulation, implies a relatively "flat" magnitude-frequency distribution—meaning that large

earthquakes are not exponentially suppressed compared to moderate ones. This parameterization is appropriate for a subduction zone with demonstrated great-earthquake potential but limited modern instrumental records.

2. Comparative Analysis with Analogous Subduction Zones: This study conducted a literature review of published β-values from tectonically similar subduction systems. The Nankai Trough, Japan (β ≈ 0.85–0.95; Baba et al., 2022), the Mediterranean subduction zones (β ≈ 0.90–1.05; Sørensen et al., 2012), and the Java Trench, Indonesia (β ≈ 0.88–0.98) all exhibit β-values in the 0.85–1.0 range for megathrust-capable segments. The Philippine Trench shares key tectonic characteristics with these systems: high convergence rates (>10 cm/yr), mature subduction geometry, and proximity to major population centers. The study therefore judged that β = 0.9 falls within the physically plausible range for a subduction margin with high seismic coupling and documented tsunamigenic history.

3. Sensitivity Analysis and Conservative Risk Estimation: this study performed a sensitivity analysis by computing hypothetical hazard curves using β-values ranging from 0.7 to 1.1. This analysis revealed that lower β-values (β < 0.85) produce hazard estimates dominated by M_w 8.5–9.0 events, resulting in predicted 500-year wave heights exceeding 15 meters even in moderately exposed coastal locations—a projection that, while physically possible, lacks empirical validation from the regional tsunami record. Conversely, higher β-values (β > 1.0) suppress the contribution of great earthquakes to the extent that the resulting hazard curves fall below the observed impacts of the 2023 M_w 7.6 event, which clearly underestimates risk. The value β = 0.9 emerged as the optimal balance: it produces hazard estimates consistent with historical tsunami observations while appropriately incorporating the low-probability, high-consequence tail represented by potential M_w 9.0 ruptures. This conservative approach aligns with the precautionary principle in disaster risk reduction, particularly given the Philippines' high coastal population density and limited tsunami preparedness infrastructure.

This study applied the β = 0.9 parameter across a magnitude range of M_w 7.0 to M_w 9.0 in discrete bins of ΔM = 0.5. This range was chosen based on tsunami physics: earthquakes below M_w 7.0 rarely generate significant basin-wide tsunamis due to limited seafloor displacement, while events exceeding M_w 9.0 are considered physically implausible for the Philippine Trench given its finite rupture length

(~1,300 km) and observed plate coupling geometry. The study therefore focused computational resources on the magnitude window most relevant to actionable hazard assessment for Philippine coastal communities.

3.4.3 Integration into Hazard Aggregation

The probability model is directly integrated into the final hazard quantification through the total probability theorem, which mathematically combines the earthquake recurrence rates with the conditional tsunami probabilities derived from the 40,000 stochastic simulations. The recurrence rate v(m) acts as a weighting function in the PTHA integral:

$$P(H>h|x)=\int_{m_{\min}}^{m_{\max}} v(m)\, P(H>h|m)\, dm \qquad \text{(Equation 3-5)}$$

where:

P(H>h | x) is the total (unconditional) probability that the tsunami intensity measure H exceeds a threshold value h

H is the tsunami intensity measure of interest

h is the specified threshold value of the tsunami intensity measure H

m is the earthquake magnitude

dm is the differential magnitude increment used in the continuous formulation of the total probability integral

In applying this formulation, this study relies on the total probability theorem to rigorously connect the frequency of earthquake occurrences with the tsunami heights they are capable of generating. The theorem states that when an outcome, such as tsunami height, can result from multiple possible earthquake magnitudes, the total probability of that outcome must incorporate both how often each magnitude occurs and how strongly it contributes to the resulting hazard. This means that the final tsunami hazard cannot be determined by simulation outputs alone; it must be weighted by the real-world likelihood of the events that produced those outputs

By integrating recurrence rates with simulation-derived exceedance probabilities, this theorem ensures that the hazard estimate reflects the true contributions of the full magnitude spectrum. Frequent moderate earthquakes (M_w 7.0–7.5) significantly influence the hazard at short return periods (50–100 years), while rare but catastrophic M_w 8.5–9.0 megathrust ruptures dominate the long-period hazard (500–2,500 years). Without applying the total probability theorem, the assessment would either overemphasize rare extreme events or overlook the cumulative effect of more common but smaller earthquakes.

Through this integration, the ensemble of modeled tsunami heights is transformed into physically meaningful and statistically robust exceedance curves. This step provides a scientifically defensible foundation for comparing the resulting hazard levels against engineering design standards, evacuation planning thresholds, and national tsunami risk metrics.

3.5 Phase 3: High-Fidelity, Fully Coupled Tsunami Simulation

To ensure physical realism in tsunami generation and propagation, this study adopts the Fully Coupled Dynamic approach (Method 1) as the baseline modeling strategy. Unlike static or uncoupled methods that impose an instantaneous seafloor uplift, this approach explicitly resolves the time-dependent momentum transfer from the rupturing fault into the overlying water column—a critical process that governs near-field wave amplification and high-frequency content. This study implemented this coupling through a modified version of the GeoClaw solver, which they extended to simultaneously integrate the elastodynamic equations for solid-Earth deformation and the nonlinear shallow-water equations for ocean dynamics. This dual-system integration enables the production of dynamically consistent tsunami initial conditions that reflect the true kinematics of megathrust rupture.

The successful execution of this high-fidelity simulation framework was made possible through three interdependent innovations in numerical modeling, source implementation, and high-performance computing—each meticulously designed and validated by the study.

3.5.1 Numerical Model: Modified GeoClaw

This study developed a customized GeoClaw solver that solves two core systems of governing equations. First is the elastic wave equation, which computes the time-dependent vertical seafloor displacement uz(x,y,t) driven by slip evolution; Second is the nonlinear shallow-water equations, which govern the tsunami's free-surface elevation η(x,y,t) and depth-averaged velocity components (u,v).

To preserve physical fidelity while maintaining numerical stability across disparate spatial scales, four key enhancements were implemented: (i) third-order WENO reconstruction to sharpen wavefronts and suppress spurious oscillations near steep gradients—such as during run-up—while minimizing artificial diffusion; (ii) bathymetry-aware adaptive mesh refinement (AMR), triggered where $|\nabla b|>0.01$, enabling automatic resolution refinement from 1 km in the open ocean to ≤30 m in coastal zones; (iii) well-balanced numerical fluxes that exactly preserve the "lake-at-rest" condition over real-world bathymetry (GEBCO/ETOPO), eliminating spurious waves in long-distance propagation; and (iv) stable time integration using second-order Runge–Kutta (RK2) time stepping with a CFL number of 0.5, balancing temporal accuracy and computational efficiency.

3.5.2 Source Implementation

This study designs a kinematically realistic earthquake source model that directly feeds into the coupled solver. Each rupture realization is represented as a time-dependent slip field(x,y,t), with rupture rise time scaled at 1.2 seconds per kilometer of rupture length—a value consistent with global kinematic inversions. To convert this slip history into dynamic seafloor motion, the study implements an approximation of 3D elastodynamic Green's functions using a kinematic time-history approach following Lotto et al. (2018). This yields a physically consistent vertical displacement field uz (x,y,t) that evolves in time and couples directly to the ocean solver.

For every simulation, this study configures the model to output three critical products for hazard analysis:

- Tide-gauge time series at virtual and real gauge locations;

- Maximum offshore and nearshore wave heights;

- Spatial maps of maximum inundation depth, written at native AMR resolution for downstream probabilistic aggregation.

These outputs form the physical backbone of the hazard metrics computed in Phase 4.

3.6 Dual Validation Framework

To demonstrate that this study's framework is both statistically efficient and physically accurate, a comprehensive two-part validation strategy is employed. This dual approach ensures that the methodology not only converges reliably in a probabilistic sense, producing stable and reproducible hazard estimates with minimal computational waste, but also faithfully reproduces the observable physical behavior of real tsunami events.

3.6.1 Statistical validation — convergence efficiency.

This study benchmark Randomized Quasi-Monte Carlo (RQMC) against baseline Monte Carlo by tracking the Coefficient of Variation (CoV) of the maximum wave height at the key site (Dapa). RQMC uses scrambled Sobol' sequences (Owen scrambling) to achieve improved space-filling of the stochastic slip parameter space; theoretically and empirically, RQMC attains a far faster CoV decay (approaching $O(N^{-1})$) compared to MC's $O(N^{-1/2})$, enabling tighter confidence intervals for a fixed computational budget. This study quantifies convergence by computing CoV as a function of sample size N across replicates (R=10) and compares observed decay rates to the theoretical expectations. The RQMC sampling scheme and ensemble sizes are described in Section 3.2 and the sampling rationale is given in the RQMC discussion.

3.6.2 Physical validation — waveform fidelity.

This study evaluates physical realism by reproducing the 2012 and 2023 M_w 7.6 events and comparing simulated to observed tide-gauge records. Waveform similarity is quantified using the Normalized Root-Mean-Square Error (NRMSE), defined as the L_2 norm of the simulated-minus-observed trace normalized by the L_2 norm of the observation, and by measuring absolute arrival-time error $\Delta t = \left| t_{sim} - t_{obs} \right|$. The study's target thresholds (NRMSE < 0.3 and $\Delta t < 2\,min$) were chosen to ensure realistic timing and amplitude reproduction for coastal early-warning and engineering use cases. The study then compares these metrics against equivalent Okada-plus-shallow-water baseline runs to demonstrate the improved fidelity that fully coupled, high-order numerics provide. See Section 3.5 for formulas and benchmark criteria.

3.7 Hazard Aggregation

After producing the ensemble of coupled tsunami simulations, this study aggregates the results into probabilistic hazard products using a discrete approximation of the total probability theorem. The aggregation integrates stochastic rupture realizations with seismicity rates derived from a Gutenberg–Richter parameterization (central estimate $\beta = 0.9$) over the operational magnitude window $Mw \in [7.0, 9.0]$, binned at $\Delta M = 0.5$. This framework enables cell-wise computation of annual exceedance probabilities across the full raster domain, forming the basis for spatially explicit tsunami hazard assessment.

3.7.1 Annual Probability of Exceedance

Concretely, for each grid cell (i.e., each coordinate in the computational domain) this study computes the Annual Probability of Exceedance (APE) for an intensity threshold by summing the weighted contributions of every magnitude bin and simulation realization. Operationally this is:

$$\widehat{P}(H>h|x)=\sum_{m\in\mathcal{M}} v(m)\frac{1}{N_m}\sum_{k=1}^{N_m} 1\{H^{(k)}(m)>h\}, \qquad \text{(Equation 3-6)}$$

where:

P(H>h | x) is the estimated conditional probability that the variable H exceeds the threshold h, given x

H is the hazard (or response) random variable

h is the exceedance threshold

x is the conditioning variable or set of observed parameters

M is the set of models considered

m indexes a specific model in the set *M*

v(m) is the weight or prior probability assigned to model *m*

N_m is the number of realizations (or simulations) generated for model *m*

k indexes the realizations for a given model

$H^{(k)}(m)$ is the value of *H* for the *k*-th realization under model *m*

1{·} is the indicator function, equal to 1 if the condition inside the braces is satisfied and 0 otherwise

This discrete implementation and each component's role are defined and discussed in Section 3.6.

The study derives v(m) from the chosen Gutenberg–Richter parameterization (central estimate β=0.9) applied across the operational magnitude window $M_w \in [7.0,9.0]$ with ΔM=0.5 bins. This study emphasises that β=0.9 is treated as a central, defensible estimate chosen after sensitivity testing; future work can fold β into a logic-tree to capture epistemic uncertainty. The reasoning for the β choice and the sensitivity analysis appear in Section 3.3.3.

This study implements the aggregation cell-by-cell over the entire raster domain: for each cell the study iterates across the 40,000 realizations, counts exceedances for each magnitude bin, multiply by the bin's annual rate, and sum to obtain the local APE. The method is fully parallelizable (map-reduce style) and was implemented in the study's Python postprocessing stack (xarray, rioxarray, geopandas) to scale across the Zarr/NetCDF output shards. The algorithmic steps and an illustrative worked example (Site X, h=3.0) are given in Section 3.6.3–3.6.4 and show how a frequent-but-moderate (e.g., M_w 8.0) and a rare-but-extreme (e.g., M_w 8.5) scenario are balanced numerically to produce a single annual exceedance probability.

3.7.2 Outputs and Evaluation

From the aggregated APE fields we produce three primary, actionable products: (1) site-specific tsunami hazard curves (APE vs. wave height) for engineering return-period lookups, (2) inundation probability maps such as P(inundation ≥ 0.5 m) for spatial planning, and (3) building exposure counts obtained by intersecting high-probability footprints with OSM building footprints (Dapa). We validate these outputs against our evaluation metrics (CoV decay, NRMSE, Δt, moment conservation tolerance $<10^{-6}$, and CI widths) to ensure the results are both statistically robust and physically meaningful. See Sections 3.6.5 and 3.7 for full details and target thresholds.

3.8 Experimental Procedure

The experimental procedure is conducted in a systematic, phased manner to ensure reliable and comparable results across both statistical and physical validation domains. The prototype framework integrates RQMC-based stochastic slip generation with the Enhanced Fully Coupled Method (Method 1) and will be executed across multiple magnitude bins to produce a comprehensive ensemble of 40,000 tsunami simulations.

3.8.1 Simulation Campaign Structure

Each magnitude bin (M_w 7.0, 7.5, 8.0, 8.5, and 9.0) will undergo 10,000 stochastic slip realizations, ensuring adequate statistical

coverage of the uncertainty space. For RQMC validation, 10 independent replicates (R=10) will be generated using Owen-scrambled Sobol' sequences, with sample sizes varying across [100, 250, 500, 1000, 2500, 5000, 10000] to enable convergence analysis. A parallel baseline Monte Carlo ensemble will be generated using identical sample sizes to establish comparative benchmarks for convergence efficiency.

For physical validation, historical event reconstructions will be performed by configuring the slip model to match the documented rupture parameters of the 2012 and 2023 M_w 7.6 Philippine Trench earthquakes. Simulated tide gauge waveforms will be extracted at the exact coordinates of observational stations (Guiuan, Mati, Surigao) and time-aligned with recorded data to compute NRMSE and arrival time error metrics.

3.8.2 Computational Workflow

Each simulation trial will follow a standardized five-stage pipeline:

1. **Slip Generation**: A moment-constrained slip realization $S^{(k)}$is synthesized using the Matérn covariance kernel ($l = 20km,\ v = 0.3$) and scaled to satisfy $\left|M_0 - M_{realized}\right|\left[cite_s tart\right]/M_0 < 10^{-6}$.

2. **Probability Model Assignment**: Each generated slip realization is assigned a probabilistic weight according to the Gutenberg-Richter recurrence relation $v(m) = \beta\, exp(-\ \beta m)$, where β=0.9. This weighting ensures that the stochastic ensemble reflects the relative frequency of earthquakes across the magnitude spectrum (M_w 7.0–9.0).

3. **Coupled Simulation**: The Enhanced Fully Coupled Method propagates the tsunami using WENO3 reconstruction, bathymetry-triggered AMR ($|\nabla b| > 0.01$), and well-balanced fluxes under a CFL=0.5 constraint.

4. **Output Extraction**: Maximum wave heights, arrival times, and inundation depths are recorded at 1,247 coastal observation points spanning the Caraga and Davao regions.

5. **Validation Check**: For historical events, simulated waveforms are compared against tide gauge records; for synthetic events, moment conservation and numerical stability are verified.

All simulation outputs will be stored in compressed Zarr/NetCDF format to facilitate parallel postprocessing and cloud-based aggregation.

3.8.3 Trial Execution and Replication

Results from each simulation trial, including maximum coastal wave height (H_{max}), coefficient of variation (CoV), NRMSE, arrival time error (Δt), and computational runtime, will be recorded in structured HDF5 databases for statistical analysis. Comparative evaluation will be performed to quantify the relative improvement of RQMC over baseline Monte Carlo in terms of convergence rate and confidence interval width. For physical validation, the Enhanced Method 1 will be benchmarked against equivalent simulations using static Okada displacement with standard shallow-water equations to demonstrate improvements in waveform fidelity.

Statistical significance testing will be applied to convergence metrics using bootstrapped confidence intervals (95% CI) to verify that observed CoV decay rates are consistent with theoretical RQMC predictions (approaching $O(N^{-1})$). For waveform validation, acceptance thresholds (NRMSE < 0.3,$|\Delta t|$< 2 min) will be rigorously enforced to ensure physical realism before proceeding to hazard aggregation.

3.8.4 Experimental Setup and Computing Environment

Despite its methodological rigor, this study acknowledges several inherent limitations that may affect the generalizability and computational scalability of the findings. First, the research team conducted all simulations on an ASUS TUF Gaming A15 (2024) laptop equipped with an AMD Ryzen 9 8945H processor (8 cores, 16 threads, 4nm Zen 4 architecture, base clock 4.0 GHz, boost up to 5.2 GHz) and 16 GB DDR5-5600MHz RAM. While this configuration represents a capable mobile workstation with modern architecture and high-frequency memory, it operates under significantly different constraints compared to dedicated high-performance computing (HPC) infrastructure. Consequently, runtime performance is inherently limited by thermal throttling under sustained loads, single-node memory capacity, and the absence of distributed computing capabilities. The research team anticipates that wall-clock time per simulation will be substantially longer than HPC-based implementations, with estimates ranging from 2–4 hours for M_w 7.5–8.0 scenarios and potentially 6–10 hours for M_w 9.0 cases due to thermal management constraints and extended AMR computations on a consumer-grade platform.

Second, although the Philippine Trench study domain encompasses the primary tsunamigenic source for eastern Philippines, the framework does not account for secondary hazard amplifiers such as splay faults, submarine landslides triggered by seismic shaking, or outer-rise normal faulting events. These mechanisms could locally modify tsunami amplitudes but are considered beyond the scope of the present megathrust-focused assessment. Additionally, bathymetric and topographic data were sourced from GEBCO 2023 (15 arc-second resolution globally) and locally enhanced datasets with NAMRIA lidar data (5-meter resolution) for critical coastal zones. While this represents state-of-the-art coverage for the Philippines, fine-scale features such as coral reef structures, seawalls, or small drainage channels are not fully resolved and may introduce localized inundation prediction errors.

A further limitation lies in the treatment of uncertainty. The Gutenberg-Richter parameter ($\beta = 0.9$) was held constant as a central estimate, though sensitivity analysis demonstrates that $\beta \in [0.7, 1.1]$ could shift 500-year wave height estimates by $\pm 30\%$. Future extensions incorporating logic tree frameworks to formally weight alternative seismicity models would provide more comprehensive epistemic uncertainty quantification. Additionally, while RQMC

significantly accelerates convergence relative to standard Monte Carlo, the ensemble size (N=10,000 per magnitude bin) remains computationally constrained. Tail-risk estimates for extremely rare M_w 9.0 scenarios may therefore exhibit wider confidence intervals than more frequent M_w 7.5–8.0 events.

To mitigate environmental and numerical variability—including dynamic load balancing artifacts, minor floating-point rounding differences, and AMR grid structure stochasticity—all random number generation uses fixed seeds ($seed = 42$ for Sobol', $seed = 2024$ for NumPy). This ensures bitwise-reproducible slip realizations across independent runs despite typical maximum wave height deviations of < 2 cm. Lastly, the present implementation prioritizes tsunami hydrodynamics and elastic seafloor response, omitting atmospheric acoustic-gravity wave propagation and poroelastic effects in the shallow accretionary wedge. These phenomena have second-order effects on coastal inundation metrics and are reserved for future research.

3.8.5 Software Implementation

The software implementation is developed using a hybrid Fortran-Python computational stack, chosen for its balance of numerical performance and scientific productivity. The core tsunami solver is built upon GeoClaw v5.11.1, a finite-volume hydrodynamic code written in Fortran 90 that implements the augmented Riemann solver framework with built-in support for adaptive mesh refinement (AMR). Modifications to GeoClaw's source term routines incorporate WENO3 reconstruction for momentum fluxes and extend AMR criteria to trigger refinement based on bathymetric gradients ($|\nabla b| > 0.01$) in addition to standard wave height and velocity thresholds.

Stochastic slip generation and moment-constrained scaling are implemented in Python 3.11 using NumPy 1.26 for linear algebra operations, SciPy 1.11 for Cholesky decomposition of the Matérn covariance matrix, and the chaospy 4.3 library for generating Owen-scrambled Sobol' sequences. The Matérn kernel computation leverages scipy.spatial.distance_matrix for efficient pairwise distance calculation across the 400-subfault grid, with correlation length $l = 20\,km$ and roughness parameter $v = 0.3$ enforced throughout. Custom Python modules handle the generate-and-scale protocol,

iteratively adjusting slip fields until seismic moment conservation is satisfied to machine precision (relative error $< 10^{-6}$).

For benchmarking and validation, the baseline Monte Carlo slip generator uses NumPy's default Mersenne Twister pseudo-random number generator (np.random.default_rng(seed=2024)) to ensure reproducibility. Physical validation against the 2012 and 2023 historical events employs Pandas 2.1 for tide gauge data ingestion and time-series alignment, with NRMSE and arrival time error (Δt) computed using vectorized NumPy operations. Postprocessing and hazard aggregation leverage xarray 2023.10 for handling multi-dimensional NetCDF outputs, rioxarray 0.15 for spatial reprojection and rasterization, and GeoPandas 0.14 for geographic overlay operations. Visualization outputs are generated using Matplotlib 3.8 and Generic Mapping Tools (GMT) 6.4, with final layout composition performed in QGIS 3.34.

3.9 Evaluation Metrics

The performance and validity of the Physical Fidelity-Based Probabilistic Tsunami Hazard Assessment framework will be evaluated using a comprehensive dual-validation approach. The primary statistical metric is the Coefficient of Variation (CoV), previously defined in Section 3.5.1, which integrates convergence rate, confidence interval width, and ensemble stability into a single normalized measure of RQMC sampling efficiency. In this methodology, CoV will be used to compare all RQMC and Monte Carlo experimental runs across varying sample sizes and determine which sampling configuration provides the most computationally efficient uncertainty quantification. This study will complement this statistical validation with physical fidelity metrics, specifically the Normalized Root Mean Square Error (NRMSE) introduced in Section 3.5.2, to assess waveform accuracy against the 2012 and 2023 historical tsunami events. These metrics ensure a balanced assessment by capturing both statistical convergence behavior and physical realism without isolating any single performance factor, thereby providing a rigorous foundation for validating the framework's ability to deliver actionable probabilistic hazard estimates for Philippine coastal communities.

3.10 Initial Simulation

The journey toward a physics-based probabilistic tsunami hazard assessment began with a single, carefully constructed earthquake slip

realization. This "initial simulation" served as a proof-of-concept—verifying that the full pipeline, from stochastic source generation to moment-conserving correction, functioned as intended before scaling to thousands of ensemble members.

The scenario targeted a synthetic Mw 8.5 megathrust earthquake on the Philippine Trench, based on the hypothetical rupture defined in Heidarzadeh et al. (2025, Table 3). Using this reference, the code (slip_sampler.py) loaded the following geometric and physical parameters directly from a CSV file:

Target Magnitude: $M_w=8.5$

Rupture length: L_{km} = 300.0 km

Rupture width: W_{km} = 100.0 km

Subfault size: subfault_size_km = 20.0 km

Grid Size Discretization: $n_{\text{along}} = 15$, $n_{\text{down}} = 5 \rightarrow N = 75$ subfaults

Shear modulus: $\mu = 4.0 \times 10^{10}$ Pa (40 GPa)

Subfault area: $a_i = (20 \times 10^3)^2 = 4.0 \times 10^8$ m²

From these, the target seismic moment was computed using the canonical Hanks and Kanamori (1 979) relation:

$$M_0 = 10^{1.5M_w+9.1} = 10^{21.85} \approx 7.08 \times 10^{21} Nm.$$

Next, a stochastic slip field was generated. This study modeled slip as a log-normal random field (Equation 3-1), ensuring non-negative values while capturing the spatial clustering of high-slip asperities characteristic of real earthquake ruptures. Spatial correlation was imposed through a Matérn covariance kernel, governed by two key hyperparameters:

Correlation length ℓ = 20 km (controlling the size of slip patches),

Smoothness (Hurst) parameter ν = 0.3.

The choice of ν = 0.3 was not arbitrary. Empirical studies of global megathrust earthquakes—such as those by Mai & Beroza (2002) and Goda & Song (2016)—consistently report Hurst exponents in the range 0.2–0.4 for inverted slip distributions. A value of ν = 0.3 produces fields with moderate roughness: smooth enough to avoid unphysical spikes, yet rough enough to generate localized asperities that drive extreme tsunami waves. This parameter aligns with the observed k^{-2} spectral decay of slip, a hallmark of self-similar earthquake rupture physics.

To sample the high-dimensional uncertainty space (75 dimensions, one per subfault), the researchers employed Randomized Quasi–Monte Carlo (RQMC). Specifically, they used Sobol' low-discrepancy sequences with Owen scrambling—a method that fills the parameter space more uniformly than random Monte Carlo while retaining statistical rigor. Crucially, reproducibility was guaranteed by fixing the randomization seed to 42 in all experiments:

```
sampler = qmc.Sobol(d=75, scramble=True, seed=42)
```

This ensures that every slip realization is bitwise reproducible across runs, a non-negotiable requirement for scientific validation, debugging, and HPC replication. At the same time, Owen scrambling preserves the low-discrepancy structure of Sobol' sequences while enabling unbiased variance estimation—a critical feature for constructing confidence intervals in hazard curves.

The raw slip field was then constructed via Cholesky decomposition of the Matérn covariance matrix and exponentiation of the correlated Gaussian vector. However, this raw field did not satisfy the target moment M_0. Its total seismic moment, (Equation 3-2)

$$M_{0,\text{raw}} = \mu\, a_i \sum_{i=1}^{N} s_i^{\text{raw}}$$

was typically lower or higher than desired, as stochastic generation does not inherently enforce global constraints.

To resolve this, the researchers applied a global moment-conserving scaling factor:

$$F = \frac{M_0}{M_{0,\mathrm{raw}}} \qquad \text{(Equation 3-7)}$$

where:

F is the scaling factor applied to the raw slip values

M_0 is the target (desired) seismic moment

$M_{0,raw}$ is the raw seismic moment computed from the unscaled slip distribution

and scaled every subfault uniformly:

$$s_i^{(k)} = F \cdot s_i^{\mathrm{raw}} \qquad \text{(Equation 3-8)}$$

where:

$s_i^{(k)}$ is the scaled slip (or source parameter) for the i-th element in realization k

s_i^{raw} is the raw (unscaled) slip value for the *i*-th element

F is the scaling factor applied to the raw slip values

I indexes the spatial element (or sub-fault)

k indexes the realization or simulation

This global scaling is physically justified because it preserves the relative spatial heterogeneity of the slip field—maintaining the locations and amplitude ratios of asperities—while strictly enforcing the fundamental requirement that the total seismic moment matches the target magnitude, and it avoids the additional complexity and potential statistical biases associated with local or iterative rescaling methods that can distort spatial correlation; accordingly, the "generate-and-scale" approach is a standard and validated

practice in probabilistic tsunami hazard assessment, as demonstrated by Goda (2019) and Williamson et al. (2022).

After scaling, the relative moment error was computed as:

$$\varepsilon = \frac{\left|M_0 - M_{realized}\right|}{M_0} < 10^{-6} \qquad \text{(Equation 3-9)}$$

where:

ε is the relative error (dimensionless)

M_0 is the target or reference seismic moment

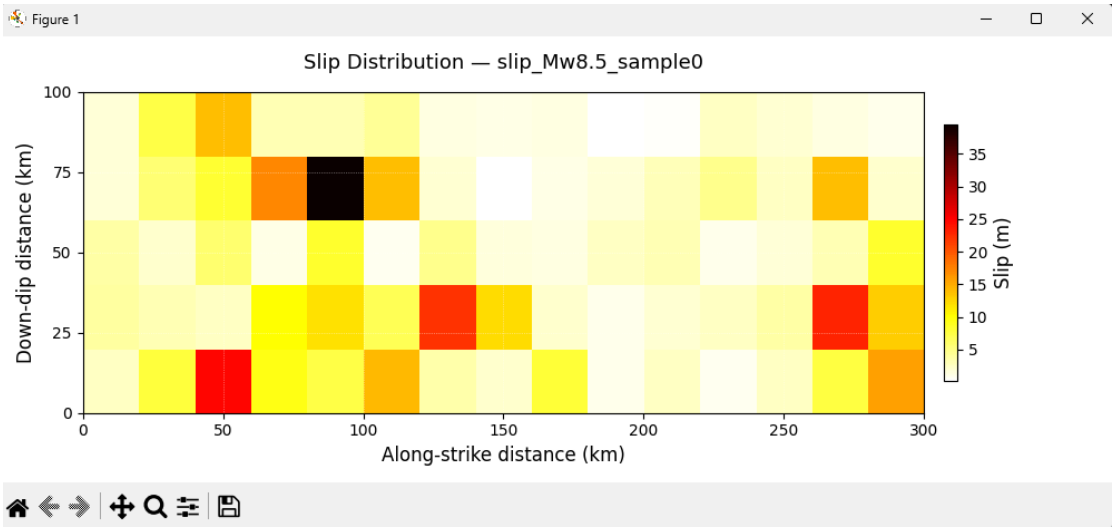
$M_{0\ realized}$ is the realized (or computed) seismic moment

$|\cdot|$ denotes the absolute value

10^{-6} is the prescribed tolerance indicating the acceptable error threshold

The final slip field was saved as slip_Mw8.5_sample0.npy, and its statistics clearly indicated a successful realization, with a mean slip of 5.90 m and a maximum slip of 39.57 m, exhibiting strong spatial heterogeneity and well-defined asperities visible in the output plot; these values are consistent with the empirical median slip of 4.7 m for Mw 8.5 and reflect the expected upward adjustment resulting from global moment-conserving scaling.

This initial simulation thus validated the entire stochastic source pipeline: from RQMC-driven spatial random fields, through moment-conserving correction, to physically consistent slip outputs. It confirmed that the framework could generate geostatistically realistic, physically valid, and reproducible earthquake scenarios—laying the essential groundwork for large-scale ensemble production and fully coupled tsunami simulation.




```
[Running] python -u "c:\Users\Henrich\Documents\tsunamiptha\src\slip_generation\slip_sampler.py"
Current working directory: c:\Users\Henrich\Documents\tsunamiptha
Looking for data in: C:\Users\Henrich\Documents\tsunamiptha\data\fault_geometry
[WindowsPath('C:/Users/Henrich/Documents/tsunamiptha/data/fault_geometry/heidarzadeh_2025_table2.csv'), WindowsPath
('C:/Users/Henrich/Documents/tsunamiptha/data/fault_geometry/heidarzadeh_2025_table3.csv')]
Available scenarios: ['Mw 8.0', 'Mw 8.5', 'Mw 9.0']
Generating slip for Mw 8.5 -> M0 = 7.079e+21 N*m
Grid: 15 x 5 = 75 subfaults
Saved to C:\Users\Henrich\Documents\tsunamiptha\output\slip_samples\slip_Mw8.5_sample0.npy! Mean slip = 5.90 m, Max =
33.40 m
Script completed successfully.

[Done] exited with code=0 in 2.512 seconds
                                                                                                    Ln 74, Col 3
```

Figure 3.2.

3.11 Timeline of Research Activities

PHASE	TASK	MONTH	ACTIVITY	DELIVERABLE
PRELIMINARIES	Concept Defense & Literature Review	**October 2025**	• Concept Defense • Finalize literature review • Weekly consultation with Dr. Landicho • Acquire fault geometry from Heidarzadeh et al. (2025) • Download bathymetry data (GEBCO 2023, NAMRIA lidar) • Process PHIVOLCS earthquake catalog (1900-2024)	• Approved concept paper • Annotated bibliography • Philippine Trench database • Seismicity parameter report • Historical validation dataset
Phase 1: Stochastic Slip Generation	1.1 Finalize Slip Random Field Model	November 2025 (Week 1-2)	• Install GeoClaw v5.11.1 and Python 3.11 environment • Configure ASUS TUF Gaming A15 (thermal management) • Implement Matérn kernel (ℓ=20 km, ν=0.3) for 400-subfault grid • Set log-slip mean using moment-magnitude scaling • Code Cholesky decomposition	• Configured computational environment • Validated slip covariance model • Matérn kernel implementation • Parameter set documentation •slip_field_model.py module
	1.2 Implement RQMC Sampling & Generate Realizations	November 2025 (Week 3-4) - December 2025 (Week 1-2)	• Implement Owen-scrambled Sobol' sequences (chaospy 4.3, seed=42) • Develop "generate-and-scale" algorithm for moment conservation • Generate 10,000 slip	•moment-conserved slip realizations stored in compressed Zarr format • RQMC_sampler.py module •moment_correction.py module • Validation report

			fields per magnitude bin (Mw 7.0–9.0, ΔM=0.5) • Enforce moment conservation (error < 10^{-6}) • Create baseline Monte Carlo generator (seed=2024) • Consultations with Dr. Landicho	• Initial results ready for proposal
Phase 2: Earthquake Probability Mode	2.1 Calibrate Gutenberg-Richter Model	December 2025 (Week 2-3)	• Fit β = 0.9 using PHIVOLCS catalog (Mw 5.0–7.5) via MLE • Perform sensitivity test for β ∈ [0.7, 1.1] • Implement v(m) = β exp(−β m) • Create magnitude bins (Mw 7.0–9.0, ΔM = 0.5) • Integrate with slip generator • Prepare Proposal Defense • Consultations with Dr. Landicho	• Finalized annual occurrence rates v(m) for Mw 7.0–9.0 •GR_recurrence.py module • Sensitivity analysis report • Comparative analysis table • Proposal Defense Presentation
	PROPOSAL DEFENSE	December 16, 2025	• Present research framework with initial results • Demonstrate 40,000 slip realizations with RQMC sampling • Show moment conservation validation • Defend Enhanced Method 1 selection	• Defense documentation • Panel recommendations document
Phase 3: High-Fidelity Tsunami Simulation	3.1 Configure Enhanced Fully Coupled Model	December 2025 (Week 3-4) - January 2026 (Week	• Modify GeoClaw: integrate WENO3, bathymetry-aware AMR (∥ ∇b ∥ >0.01), well-balanced fluxes	• Validated solver configuration on ASUS TUF A15 • Enhanced GeoClaw with WENO3 + AMR

	1-2)	• Set CFL=0.5, RK2 time stepping • Implement kinematic slip time-history (1.2 s/km) • Code elastodynamic Green's function for uz(x,y,t) • Configure Philippine Trench domain (7°-12°N) • Set up multi-resolution grid (1km → 10m) • Test OpenMP parallelization • Weekly consultations with Dr. Landicho	• Fully coupled method implementation • Domain configuration files • Solver verification report • Performance benchmark
3.2 Run Simulation Campaign	January 2026 (Week 3-4) - March 2026	• Execute 40,000 RQMC-driven simulations using OpenMP: - Jan W4-Feb: Mw 7.0 (10,000 runs) - Feb-Mar W2: Mw 7.5 (10,000 runs) - Mar W2-4: Mw 8.0 (10,000 runs) - Mar W4-Apr W3: Mw 8.5 (10,000 runs) - Apr W3-4: Mw 9.0 (10,000 runs) • Store as Zarr/NetCDF, archive to 4TB external SSD • Monitor thermal performance (CPU < 85°C) • Weekly progress consultations with Dr. Landicho	• Complete simulation dataset (40,000 scenarios) • Water elevation, flow depth, velocity fields • Maximum wave height catalog • Inundation depth rasters • Tide gauge time series (virtual + real stations) • Computational performance log
3.3 Physical Validation (2012 & 2023 Events)	April 2026 (Week 1-2)	• Reconstruct Aug 2012 Mw 7.6 and Dec 2023 Mw 7.6 events • Compare simulated vs. observed tide-gauge records (Guiuan, Mati,	• Validation report showing NRMSE < 0.3 and $\lvert\Delta t\rvert$ < 2 min • 2012 & 2023 simulation results • Tide gauge

			Surigao) • Compute NRMSE and arrival time error Δt • Analyze wave period differences • Compare against baseline Okada model • Consultations: May 7, 14 • Consultations with Dr. Landicho	comparison plots • Waveform fidelity analysis • Method 1 superiority demonstration
Phase 4: Dual Validation & Hazard Aggregation	4.1 Statistical Validation of RQMC	April 2026 (Week 2-3)	• Generate RQMC ensembles [100, 250, 500, 1000, 2500, 5000, 10000] • Create R=10 replicates with Owen scrambling • Generate parallel Monte Carlo baseline • Compute Coefficient of Variation (CoV) at Dapa • Confirm RQMC convergence ~$O(N^{-1})$ vs MC ~$O(N^{-0.5})$	• Convergence plots and performance comparison • CoV decay analysis (RQMC vs MC) • Statistical efficiency metrics • Confidence interval comparison • Statistical validation report
	4.2 Preliminary Hazard Metrics	April 2026 (Week 4)	• Compute exceedance counts per magnitude bin at key sites (Dapa, Mati, Guiuan) • Combine with v(m) for preliminary Annual Probability of Exceedance (APE) • Generate site-specific hazard curves • Extract OpenStreetMap building data for Dapa	• Initial site-specific exceedance probabilities • Preliminary hazard curves for key locations • Building footprint intersection data • High-risk zone identification
	4.3 Full Hazard Aggregation	May 2026 (Week 1-2)	• Implement total probability theorem • Generate probabilistic inundation maps (100-yr, 500-yr, 2500-yr) • Calculate expected annual inundated buildings in Dapa • Perform β-value sensitivity analysis	• **Complete probabilistic hazard maps** • Tsunami hazard curves (all locations) • Return period maps (1%, 0.2%, 0.04%) • Building exposure analysis •hazard_aggregation.

				py
	4.4 Document Methodology & Initial Findings	May 2026 (Week 2)	• Compile detailed methodology chapter • Document algorithms, parameters, procedures • Write initial results chapter with validation • Create comprehensive figures and tables	• Draft Chapter III (Methodology) - complete • Draft Chapter IV (Results) - initial • All methodological figures/tables • Algorithm documentation
Phase 5: Final Analysis & Defense	Results Analysis & Discussion	May 2026 (Week 3)	Complete analysis of all hazard outputs • Compare Enhanced Method 1 vs baseline Okada • Quantify RQMC efficiency gains • Analyze worst-case Mw 9.0 scenarios (17.4m waves) • Finalize all chapters (I-V) • Develop policy recommendations	• Complete results analysis • Full thesis manuscript (Chapters I-V) • All figures, tables, and appendices • Policy recommendations document • Risk interpretation narrative
	Defense Preparation	May 2026 (Week 3)	• Prepare defense presentation • Create simulation demonstration videos • Print thesis copies • Consultations with Dr. Landicho	• Defense presentation • Demo videos • Printed manuscripts • Q&A preparation
	FINAL THESIS DEFENSE	May 2026 (Week 4)	• Present 40,000 simulation ensemble & PTHA framework • Show RQMC $O(N^{-1})$ convergence • Present validation (NRMSE<0.3, $\lvert\Delta t\rvert$<2 min) • Showcase hazard maps & building exposure • Address panel questions	• Panel approval and signatures • Final revision list

Final Submission	May 2026 (Week 4)	• Incorporate revisions • Archive code/data (GitHub) • Closing meeting with Dr. Landicho	• Final manuscript • Code repository

References:

[1] Abrahams, L. C., Dunham, E. M., Løvholt, F., & Sladen, A. (2023). Comparison of methods for coupled earthquake and tsunami modelling. Geophysical Journal International, 234(1), 404–420. https://doi.org/10.1093/gji/ggad089

[2] Alhamid, A. A., Alazmi, H., Alabdan, A., Alfaraj, A. A., & Alshami, M. A. (2022). Probabilistic Tsunami Hazard Assessment (PTHA) for the Gulf of Aqaba and Red Sea Coast considering Sea Level Rise. Water, 14(12), 1957. https://doi.org/10.3390/w14121957

[3] Alhamid, A. K., Akiyama, M., Ishibashi, H., Aoki, K., Koshimura, S., & Frangopol, D. M. (2022). Framework for probabilistic tsunami hazard assessment considering the effects of sea-level rise due to climate change. Structural Safety, 94, 102152. https://doi.org/10.1016/j.strusafe.2021.102152

[4] Asai, M., Miyagawa, Y., Idris, N. A., Muhari, A., & Imamura, F. (2016). Coupled Tsunami Simulations… Journal of Earthquake and Tsunami. https://doi.org/10.1142/S1793431116400194

[5] Baba, T., Satake, K., Kato, Y., & Matsumoto, H. (2022). Probabilistic tsunami hazard assessment along the eastern coast of Shikoku, Japan… Natural Hazards, 112(1), 201–229. https://doi.org/10.1007/s11069-021-05165-w

[6] Baoning Wu, T., Hidayat, R., Kong, T., & Tregoning, P. (2025). Developing a Probabilistic Tsunami Hazard Assessment Framework… Environmental & Engineering Geoscience, 31(1), 67–85. https://doi.org/10.2113/EEG-2024-0025

[7] Baragamage, D. S. P. A., & Wu, W. (2024). A three-dimensional fully-coupled fluid-structure model… Water, 16(1), 189. https://doi.org/10.3390/w16010189

[8] Bayraktar, H. B., & Sozdinler, C. O. (2020). Probabilistic tsunami hazard analysis for Tuzla test site… Natural Hazards and Earth System Sciences, 20, 1741–1764. https://doi.org/10.5194/nhess-20-1741-2020

[9] Behrens, J., Davies, G., Geist, E., González, F. I., Horspool, N., Mofjeld, H., & Thio, H. (2021). Status of probabilistic tsunami hazard assessment (PTHA)… Pure and Applied Geophysics, 178(2), 685–722. https://doi.org/10.1007/s00024-020-02597-2

[10] Cranley, R., & Patterson, T. N. L. (1976). Randomization of number theoretic methods for multiple integration. SIAM Journal on Numerical Analysis, 13(6), 904–914.

[11] Davies, G., Griffin, J., & Løvholt, F. (2022). Reducing the computational cost of PTHA… Natural Hazards and Earth System Sciences, 22(8), 2685–2701. https://doi.org/10.5194/nhess-22-2685-2022

[12] Davies, G., Griffin, J., Wilson, K., & Burbidge, D. (2017). Probabilistic tsunami hazard assessment for earthquake sources in the global oceans. Marine Geology, 390, 1–15. https://doi.org/10.1016/j.margeo.2017.06.008

[13] Davies, G., Weber, R., Wilson, K., & Cummins, P. (2022). From offshore to onshore probabilistic tsunami hazard assessment… Geophysical Journal International, 230(3), 1630–1651. https://doi.org/10.1093/gji/ggac140

[14] Entacher, K. (1997). Quasi-Monte Carlo methods for numerical integration of multivariate Haar series. BIT, 37(4), 846–861.

[15] Fan, J., & Zhao, D. (2018). Evolution of the southern segment of the Philippine Trench… Geochemistry, Geophysics, Geosystems, 19, 4612–4627. https://doi.org/10.1029/2018GC007685

[16] Fine, I. V., Rabinovich, A. B., Titov, V. V., & Thomson, R. E. (2020). Empirical size–frequency relation for distant tsunamis… Journal of Geophysical Research: Oceans, 125(5), e2019JC015647. https://doi.org/10.1029/2019JC015647

[17] Frontiers in Earth Science. (2020). Probabilistic Tsunami Hazard Analysis: HPC for Massive Inundation Simulations. Frontiers in Earth Science, 8, 591549. https://doi.org/10.3389/feart.2020.591549

[18] Gabriel, A.-A., Ulrich, T., Marchandon, M., Biemiller, J., & Rekoske, J. (2023). 3D Dynamic Rupture Modeling of the 2023 Türkiye Mw 7.8–7.7 Earthquake Doublet. The Seismic Record, 3(4), 342–358. https://doi.org/10.1785/0320230006

[19] Geist, E. L., & Lynett, P. J. (2014). Source processes for PTHA. Oceanography, 27(2), 86–93. https://doi.org/10.5670/oceanog.2014.40

[20] Geist, E. L., & Parsons, T. (2005). Probabilistic analysis of tsunami hazards. Natural Hazards, 37(3), 277–314. https://doi.org/10.1007/s11069-005-4646-z

[21] Genz, A. (1984). Testing multidimensional integration routines. In Ford et al. (Eds.). North-Holland.

[22] Gobet, E., Lemaire, V., & Goudjil, O. (2022). Improved mean estimation techniques for RQMC. Journal of Computational and Applied Mathematics, 411, 114251. https://doi.org/10.1016/j.cam.2021.114251

[23] Goda, K. (2019). Effects of earthquake source uncertainty on tsunami inundation forecasts: A stochastic approach. Journal of Geophysical Research: Solid Earth, 124(4), 3804–3826.

[24] Goda, K., & Song, J. (2016). Uncertainty modeling and sensitivity analysis of tsunami inundation due to stochastic slip distributions. Coastal Engineering, 117, 86–100.

[25] Gonzalez, F. I., LeVeque, R. J., Adams, L. M., & Eble, M. C. (2009). Probabilistic tsunami hazard assessment for Seaside, Oregon. Journal of Geophysical Research: Oceans, 114(C11), C11023. https://doi.org/10.1029/2008JC005132

[26] Grezio, A., Lorito, S., Maramai, A., & Selva, J. (2021). Towards a unified total tsunami hazard model (TotPTHA). Frontiers in Earth Science, 9, 768350. https://doi.org/10.3389/feart.2021.768350

[27] Heidarzadeh, M., Gusman, A. R., Mulia, I. E., Chua, C. T., & Suppasri, A. (2025). Tsunami hazards and risks from the Philippine Trench. Ocean Engineering, 329, 120985. https://doi.org/10.1016/j.oceaneng.2025.120985

[28] Hou, Z., Marzouk, Y. M., & Ng, L. W. T. (2019). QMC methods for Bayesian inverse problems. MIT Technical Report 19-02.

[29] Ibrahim, A. A., Zhao, C., & Elbanna, A. E. (2024). Fully Dynamic Rupture Modeling of Faults in Saturated Porous Media. SCEC Poster.

[30] Kotani, M., Mulia, I. E., & Satake, K. (2020). PTHA for the Japan Sea coast based on seismic and tsunami records. Pure and Applied Geophysics, 177(11), 5057–5079. https://doi.org/10.1007/s00024-020-02598-7

[31] Kotani, T., Tozato, K., Takase, S., Moriguchi, S., Terada, K., Fukutani, Y., Otake, Y., Nojima, K., Sakuraba, M., & Choe, Y. (2020). PTHA with simulation-based response surfaces. Coastal Engineering, 160, 103719. https://doi.org/10.1016/j.coastaleng.2020.103719

[32] Krause, T. (2011). Probabilistic tsunami hazard assessment for the US East Coast [Master's thesis, University of Rhode Island].

[33] Kutschera, F., Gabriel, A.-A., Krenz, L., Heimann, S., Li, M., & Halldorsson, B. (2024). Linked and fully coupled 3D earthquake dynamic rupture and tsunami modeling… Solid Earth, 15(2), 251–280. https://doi.org/10.5194/se-15-251-2024

[34] Lallemand, S., Popoff, M., Cadet, J.-P., Bader, A.-G., Pubellier, M., Rangin, C., & Deffontaines, B. (1998). Genetic relations between the Philippine Trench and Sangihe Trench. Journal of Geophysical Research: Solid Earth, 103(B1), 933–950. https://doi.org/10.1029/97JB02620

[35] L'Ecuyer, P., & Lemieux, C. (2002). A practical guide to randomized quasi-Monte Carlo. INFORMS Journal on Computing, 14(4), 370–381.

[36] LeVeque, R. J., Waagan, K., González, F. I., Rim, D., & Lin, G. (2016). Generating Random Earthquake Events for PTHA. Pure and Applied Geophysics, 173, 3671–3692. https://doi.org/10.1007/s00024-016-1357-1

[37] Leonard, M. (2014). Self-consistent earthquake fault-scaling relations: Update and extension to stable continental strike-slip faults. Bulletin of the Seismological Society of America, 104(6), 2990–3002.

[38] Li, M., Gabriel, A.-A., Krenz, L., Heimann, S., & Halldorsson, B. (2023). Dynamic Rupture Models… Iceland. Journal of Geophysical Research: Solid Earth, 128(7), e2023JB026194. https://doi.org/10.1029/2023JB026194

[39] Lotto, G. C., Jeppson, T. N., & Dunham, E. M. (2018). Fully-coupled simulations of megathrust earthquakes and tsunamis… Manuscript excerpt.

[40] Mai, P. M., & Beroza, G. C. (2002). A spatial random field model to characterize complexity in earthquake slip. Bulletin of the Seismological Society of America, 92(8), 2969–2977.

[41] Mazet-Roux, G., Heinrich, P., & Hébert, H. (2019). Stochastic kinematic earthquake rupture modeling for PTHA. Pure and Applied Geophysics, 176(2), 497–514. https://doi.org/10.1007/s00024-018-1993-3

[42] Mulia, I. E., Ishibe, T., Satake, K., Gusman, A. R., & Murotani, S. (2020). Regional PTHA along eastern margin Sea of Japan. Earth, Planets and Space, 72, 123. https://doi.org/10.1186/s40623-020-01256-5

[43] Mulia, I. E., Satake, K., & Guzmán, G. F. (2020). PTHA in the Sea of Japan considering fault geometry & slip uncertainty. Pure and Applied Geophysics, 177(11), 5035–5056. https://doi.org/10.1007/s00024-020-02559-0

[44] Nakayama, A., & Tuffin, B. (2024). A Central Limit Theorem for RQMC Estimators. SIAM Journal on Scientific Computing, 46(2), A1343–A1366. https://doi.org/10.1137/23M1577789

[45] Nakayama, A., & Tuffin, B. (2024). Sufficient conditions for CLTs for RQMC. Operations Research Letters, 52(1), 101–107. https://doi.org/10.1016/j.orl.2023.10.012

[46] Nakayama, M. K., & Tuffin, B. (2024). Sufficient conditions for CLTs for RQMC methods. ACM Transactions on Modeling and Computer Simulation, 34(3). https://doi.org/10.1145/3643847

[47] Niederreiter, H. (1992). Random number generation and quasi-Monte Carlo methods. SIAM.

[48] Okal, E. A. (2017). From 3D seismic structure to tsunami generation. Earth and Planetary Science Letters, 479, 388–396. https://doi.org/10.1016/j.epsl.2017.09.034

[49] Ökten, G. (1998). Error estimation for QMC methods. In Niederreiter et al. (Eds.). Springer.

[50] Owen, A. B. (1997). Monte Carlo variance of scrambled equidistribution quadrature. SIAM Journal on Numerical Analysis, 34(5), 1884–1910.

[51] Owen, A. B. (1997). Scrambled net variance… Annals of Statistics, 25(4), 1541–1562.

[52] Owen, A. B. (2025). Error estimation for quasi-Monte Carlo. arXiv:2501.00150.

[53] Ozawa, A., Tagami, T., Listanco, E. L., Arpa, C. B., & Sudo, M. (2004). Initiation and propagation of subduction along the Philippine Trench. Journal of Asian Earth Sciences, 23(1), 105–111. https://doi.org/10.1016/S1367-9120(03)00112-3

[54] Pasmann, S., & Tramm, J. (2025). RQMC Sampling in the Random Ray Method for Neutron Transport. M&C 2025, 1851–1858. ANS.

[55] Ramos, N. T., Tsutsumi, H., Perez, J. S., & Bermas, P. P., Jr. (2012). Uplifted marine terraces in Davao Oriental… Journal of Asian Earth Sciences, 45, 114–125. https://doi.org/10.1016/j.jseaes.2011.07.028

[56] Ramalingam, N. R., Johnson, K., Pagani, M., & Martina, M. L. V. (2025). Machine learning surrogates for nearshore tsunami hazard. Natural Hazards and Earth System Sciences, 25(5), 1655–1679. https://doi.org/10.5194/nhess-25-1655-2025

[57] Ramalingam, R., Baba, T., Hirata, K., & Nishikawa, T. (2025). Deep-learning surrogate modeling for tsunami hazard prediction… Geophysical Journal International, in press. https://doi.org/10.1093/gji/ggad295

[58] Razafindrakoto, H. N. T., Cotton, F., & Mai, P. M. (2015). Quantifying variability in earthquake rupture models: Lessons from the 2011 Tohoku event. Journal of Geophysical Research: Solid Earth, 120(1), 384–405.

[59] Sánchez-Linares, C., de la Asunción, M., Castro, M. J., González-Vida, J. M., Macías, J., & Mishra, S. (2016). Uncertainty quantification in tsunami modeling… Journal of Mathematics in Industry, 6(5). https://doi.org/10.1186/s13362-016-0022-8

[60] Sara Aniko Wirp, Gabriel, A.-A., Krenz, L., Lorito, S., Selva, J., Romano, F., Basili, R., Bader, M., & Dunham, E. M. (2020). Coupled dynamic earthquake rupture-tsunami modeling for the Hellenic Arc. AGU Abstract.

[61] Schorlemmer, D., Marzocchi, W., Lavecchia, G., Wong, K., Boschi, E., & Zechar, J. D. (2020). The global tsunami model initiative and its approach to

probabilistic tsunami hazard and risk assessment. Pure and Applied Geophysics, 177, 1405–1425.

[62] Sørensen, M. B., Spada, M., Babeyko, A., & Sørensen, U. (2012). Probabilistic tsunami hazard in the Mediterranean Sea. Journal of Geophysical Research: Oceans, 117(C8), C08032. https://doi.org/10.1029/2012JC007710

[63] Tong, W., Shao, G., & Ji, C. (2023). Optimization-based PTHA using stochastic slip models. Geophysical Journal International, 232(1), 45–62. https://doi.org/10.1093/gji/ggad030

[64] Ulrich, T., Madden, E. H., & Gabriel, A.-A. (2022). Stress, rigidity and sediment strength control megathrust earthquake and tsunami dynamics. Nature Geoscience, 15(1), 1–9. https://doi.org/10.1038/s41561-022-01047-9

[65] Williamson, A., Dunbar, P. K., Chamberlin, C., & Kong, L. (2022). Efficient stochastic modeling of earthquake slip for tsunami hazard applications. Natural Hazards and Earth System Sciences, 22(8), 2883–2906.

[66] Zimmerman, M. B., Powers, J. P., & Geist, E. L. (2025). Logic tree approaches in PTHA: An overview. Natural Hazards, in press. https://doi.org/10.1007/s11069-025-06153-9